

Matrices as Transformations

Unit 2-4 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 2-4 slides and follow them section by section. They go well beyond the slides: every idea comes with a plain-words explanation, several fully worked examples (easy ones first), pictures, and “Watch out” boxes for the mistakes students most often make. Every exercise is followed by a worked solution; try each one before reading on.

Contents

1	Why Transformations?	1
2	A Matrix Is a Function	4
3	Reading the Columns	13
4	The Catalogue	20
5	Length-Preserving Transformations	32
6	Composition and Order	39
7	Review	48

1 Why Transformations?

1.1 A Problem to Motivate Us

Here is a task that comes up in medical imaging:

“A scan is stored as a list of pixel coordinates, one point per pixel. The radiologist wants it turned by 30° and enlarged by half. Write the rule that moves every point, in a form a computer can apply to a million points at once.”

Each pixel’s location is a point in the plane, and we describe it by the vector from the origin to that point,

$$\bar{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The rule we want takes a vector **in** (where a pixel is now) and returns a vector **out** (where that pixel should go). A rule that turns each input into exactly one output is a **function**, the idea from Unit 1; the only change is that the inputs and outputs are now vectors instead of numbers. Unit 2-3 already told us what kind of object does that: a matrix.

So the answer will be a single 2×2 matrix A , and the whole scan moves in one product. Stack the million points side by side as the columns of a matrix X and compute AX . Each column of a product is A times the matching column of X (Unit 2-3), so each column of AX is one pixel, moved to its new place.

A scan, written as a matrix. To make this concrete, shrink the scan to three pixels. We measure every pixel's position from the point we want to turn about (the center of the scan, say), so that point is the origin; Section 2.2 explains why this choice matters. Suppose the three pixels sit at $(100, 0)$, $(0, 100)$, and $(100, 100)$. Stacked side by side as columns, their position vectors form

$$X = \begin{bmatrix} 100 & 0 & 100 \\ 0 & 100 & 100 \end{bmatrix},$$

a 2×3 matrix: two rows because each pixel has two coordinates, three columns because there are three pixels. A real scan is exactly the same object with a million columns. Once we have found the matrix A , the product AX will be another 2×3 matrix whose columns are the three pixels in their new places. We finish this computation in Section 4.2.

Why insist on a single matrix rather than a rule applied pixel by pixel? Because computers are built to multiply matrices quickly. Graphics cards and the numerical libraries data scientists use (NumPy among them) are tuned for exactly this operation, so one product $A @ X$ moves a million pixels far faster than a loop that moves them one at a time. The same is true of any dataset stored as a matrix.

By the end of these notes you will be able to:

- **write down** the matrix for a turn, a stretch, a flip, or a squash, just by looking at where two arrows land;
- **read** what any 2×2 matrix does in the same way;
- **chain** several moves into one matrix.

1.2 Where This Shows Up in Data Science

The product $A\bar{x}$ has a geometric reading, a picture of what it does to space, and that picture is everywhere:

- **Imaging and graphics.** Rotating, scaling, and flipping images is matrix multiplication. Image classifiers are trained on rotated and flipped copies of every picture, so that they learn to recognize an object however it is turned. This practice is called **augmentation**, and all of those copies are produced this way.
- **PCA is a rotation.** Principal component analysis, the capstone of this unit, turns the axes so that they line up with the data. A rotation is a matrix, and in these notes we build it.
- **Change of basis is a matrix.** Converting coordinates from one basis to another, the Unit 2-2 idea, is one matrix-vector product.

- A **neural network layer** is a matrix transformation followed by a simple nonlinear step. Understanding what a matrix does to space is understanding what a layer does to data. For example, a layer that reads three measurements of a patient (age, systolic blood pressure, cholesterol) and produces two numbers is a 2×3 matrix: three numbers in, two out.

Two of these deserve a closer look, because they are everyday work in a medical setting.

Registration. A patient has an MRI in January and a follow-up MRI in June. To see whether a tumor has grown, the two scans must first be lined up, because nobody lies in the scanner in exactly the same position twice. Lining them up is called **registration**, and most of it is finding the turn and the rescaling that carry one scan onto the other: a matrix like the one in our motivating problem. (Registration usually needs a sideways slide as well; Section 2.2 explains why that part is handled separately.)

Augmentation. A photo of a skin lesion shows the same lesion whether it is turned a quarter turn or mirrored, so a classifier for skin lesions can be trained on turned and mirrored copies of every photo. That multiplies the training data at no cost, and each copy is one matrix applied to the pixel coordinates.

Watch out

Not every transformation is harmless for every kind of data. A chest X-ray mirrored left to right shows the heart on the right side of the chest, which almost no real patient has. A model trained on such copies learns something false about anatomy. The matrix does exactly what it is told; deciding which transformations make sense for the data is the analyst's job.

Unit 2-3 taught us how to *compute* $A\bar{x}$. These notes ask what it **looks like**.

1.3 What We Will Build

1. A matrix as a **function** that moves every point, and what makes it **linear**.
2. The key idea: **the columns of A are where the unit vectors $\hat{i}$ and $\hat{j}$ land**. With it we can read any matrix as a picture, and build any picture as a matrix.
3. The **catalogue** of basic moves: scaling, rotation, reflection, shear, projection, and permutation.
4. Which transformations **preserve length**, and the algebraic test $A^\top A = I$ that detects them.
5. **Composition**: two moves in a row are one matrix product, and the order matters.

Sections 2 to 6 each end with a short set of extra practice problems, with full solutions, beyond the exercises from the slides.

Three notebooks go with these notes, and pointers to the relevant section of each appear throughout:

- U2-4_Transformations-1_MatrixAsMap.ipynb
- U2-4_Transformations-2_NamedTransforms.ipynb
- U2-4_Transformations-3_UnitCircleAnimation.ipynb

2 A Matrix Is a Function

2.1 One Point, Two Readings

Take

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad \bar{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

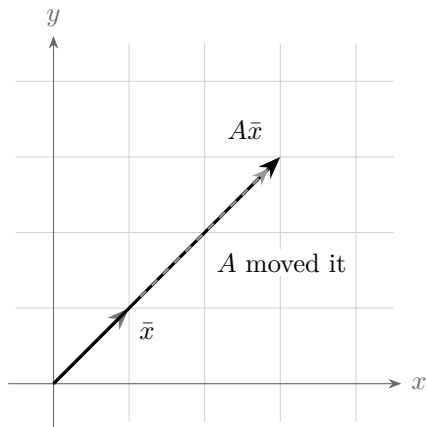
By the column reading of the matrix-vector product (Unit 2-3), $A\bar{x}$ is the first entry of $\bar{x}$ times the first column of A , plus the second entry of $\bar{x}$ times the second column:

$$A\bar{x} = 1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

The row reading from Unit 2-3 gives the same answer, as it must: each entry of $A\bar{x}$ is a row of A dotted with $\bar{x}$, so $A\bar{x} = \begin{bmatrix} 2 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. The row reading is often the quicker way to compute; the column reading is the one that will give us pictures.

There are two ways to read this one calculation.

- **Reading 1 (Unit 2-3):** the answer is a **combination of A 's columns**, one of the first column plus one of the second.
- **Reading 2 (today):** picture $\bar{x}$ as an arrow from the origin to the point $(1, 1)$. After multiplying by A , the arrow points to $(3, 3)$ instead. **A moved the point.**



The second reading changes how we think about a matrix. It stops being a table of numbers and becomes an **action**: something that happens to every point of the plane.

Definition: Linear transformation

The **linear transformation** of a matrix A is the function

$$T(\bar{x}) = A\bar{x} :$$

every vector of $\mathbb{R}^n$ goes in, a vector of $\mathbb{R}^m$ comes out, and one fixed matrix does the work for all of them.

Recall that $\mathbb{R}^n$ is the set of all vectors with n entries. An $m \times n$ matrix can only multiply a vector with n entries (one per column), and it returns a vector with m entries (one per row); that is why the inputs live in $\mathbb{R}^n$ and the outputs in $\mathbb{R}^m$. In these notes A is almost always 2×2 , so the inputs and the outputs are both points of the plane $\mathbb{R}^2$, and we can draw them.

This is a function exactly in the Unit 1 sense, with vectors in place of numbers. The matrix A is not an input: it is the *rule*, chosen once and then applied to every vector that comes in. The output of a transformation is also called the **image** of the input: the image of a vector is simply where it lands.

A function table. In Unit 1 a function could be described by a table of inputs and outputs. Here is part of the table for $T(\bar{x}) = A\bar{x}$ with the same $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. Each output is one matrix-vector product; compute two or three of them yourself as a warm-up.

input $\bar{x}$	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
output $A\bar{x}$	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 5 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -3 \end{bmatrix}$

Unlike a table in Unit 1, this one never needs to be finished or looked up: the matrix produces the output for any input in the plane, on demand.

Worked example: one point, both readings. Where does $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ send the point $(2, 1)$? By the column reading, take 2 of the first column and 1 of the second:

$$A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

By the row reading, dot each row with $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$: the first entry is $2 \cdot 2 + 1 \cdot 1 = 5$ and the second is $0 \cdot 2 + 3 \cdot 1 = 3$. Both give $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$: the point $(2, 1)$ is moved to $(5, 3)$. If your two readings ever disagree, there is an arithmetic slip somewhere, so computing both is a cheap check.

Worked example: negative entries. Take a different matrix and a point in the second quadrant,

$$M = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Column reading: -1 of the first column plus 2 of the second,

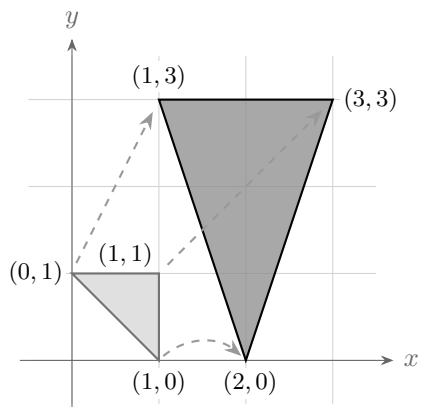
$$M\bar{x} = -1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix} + \begin{bmatrix} -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \end{bmatrix}.$$

Row reading: $1 \cdot (-1) + (-2) \cdot 2 = -5$ and $3 \cdot (-1) + 1 \cdot 2 = -1$. The point $(-1, 2)$, up and to the left of the origin, is carried to $(-5, -1)$, down and to the left. Nothing about a transformation keeps a point on its own side of an axis; the signs take care of themselves as long as you keep track of them.

Worked example: moving a whole shape. Here is the scan idea in miniature. Take the triangle with corners $(1, 0)$, $(1, 1)$, and $(0, 1)$, and move it with $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. Stack the three corners as the columns of one matrix and multiply once:

$$AX = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}.$$

Each column of AX is A times the matching column of X : $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ (the first column of A), $A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ (the example we started with), and $A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (the second column of A). So the triangle is carried to the triangle with corners $(2, 0)$, $(3, 3)$, and $(1, 3)$.



The triangle (light) and its image under $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ (dark); dashed arrows join each corner to its image.

We moved only the three corners, yet we know where the whole triangle went: as the next subsection shows, a matrix sends straight edges to straight edges, so each edge follows its two corners. That is exactly why the scan problem can be solved by moving points one column at a time.

A transformation between different spaces: tablets to doses. Not every matrix is square, and the definition allows the inputs and outputs to have different sizes. A pharmacy stocks three tablets. Tablet 1 contains 10 mg of drug A and 5 mg of drug B; tablet 2 contains 20 mg of drug A and no drug B; tablet 3 contains 25 mg of drug B and no drug A. Record each tablet's contents as a column:

$$D = \begin{bmatrix} 10 & 20 & 0 \\ 5 & 0 & 25 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{mg of drug A} \\ \leftarrow \text{mg of drug B} \end{array}$$

A patient's daily regimen is a vector $\bar{n}$ in $\mathbb{R}^3$ that counts how many of each tablet they take, and $D\bar{n}$ is their total daily dose, a vector in $\mathbb{R}^2$. For the regimen of two of tablet 1, one of tablet 2, and one of tablet 3,

$$D \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 10 \\ 5 \end{bmatrix} + 1 \begin{bmatrix} 20 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 25 \end{bmatrix} = \begin{bmatrix} 40 \\ 35 \end{bmatrix} :$$

40 mg of drug A and 35 mg of drug B per day. The column reading is exactly how a pharmacist thinks: each tablet contributes its own column, as many times as it is taken. This transformation takes vectors in $\mathbb{R}^3$ to vectors in $\mathbb{R}^2$ ($m = 2$, $n = 3$). We cannot draw it the way we draw the 2×2 examples, but it is a function on vectors all the same, and everything in the next subsection applies to it.

Notebook U2-4_Transformations-1_MatrixAsMap.ipynb, §1, the point before and after

2.2 Why “Linear”: the Two Rules and Their Consequences

Why the word *linear*? Because the transformation respects the two things we can do with vectors: add them and scale them.

The two rules of linearity

For every pair of vectors $\bar{u}, \bar{v}$ and every scalar c ,

$$A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}, \quad A(c\bar{u}) = cA\bar{u}.$$

The first rule is the distributive rule for matrix products from Unit 2-3. The second is just as direct: each entry of $A(c\bar{u})$ is a row of A dotted with $c\bar{u}$, and the factor c comes out of every term of that dot product, leaving c times the matching entry of $A\bar{u}$.

In words: move a sum, or move the parts and then add, and you get the same answer. Likewise, stretch an arrow and then move it, or move it and then stretch it, and you get the same answer. That is what **linear** means.

Worked example: checking both rules on numbers. Take $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ again, with $\bar{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\bar{v} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, and $c = 3$.

- **Add first, then move.** $\bar{u} + \bar{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, and $A\begin{bmatrix} 4 \\ 1 \end{bmatrix} = 4\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \end{bmatrix}$.
- **Move first, then add.** $A\bar{u} = 1\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ and $A\bar{v} = 3\begin{bmatrix} 2 \\ 0 \end{bmatrix} - 1\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$, so $A\bar{u} + A\bar{v} = \begin{bmatrix} 9 \\ 3 \end{bmatrix}$. The same.
- **Scale first, then move.** $3\bar{u} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$, and $A\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 6\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 18 \end{bmatrix}$.
- **Move first, then scale.** $3A\bar{u} = 3\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 12 \\ 18 \end{bmatrix}$. The same.

The practical payoff: once you know $A\bar{u}$ and $A\bar{v}$, you get $A(\bar{u} + \bar{v})$, $A(3\bar{u})$, and in fact every combination $a\bar{u} + b\bar{v}$ without multiplying by A again. Section 3 turns this into the key idea of the whole unit.

Linearity at the pharmacy. For the tablets-to-doses matrix D , the two rules are plain common sense. Doubling a regimen doubles the dose, and adding one more of tablet 3 adds that tablet's column to the dose:

$$D \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix} = 2D \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 80 \\ 70 \end{bmatrix}, \quad D \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = D \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + D \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 40 \\ 35 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \end{bmatrix} = \begin{bmatrix} 40 \\ 60 \end{bmatrix}.$$

No new multiplication is needed, because the dose is built out of the tablets and nothing else.

Three consequences follow, and you can see them in every picture in these notes.

- **The origin stays put:** $A\bar{0} = \bar{0}$. Take $c = 0$ in the second rule: $A(0\bar{u}) = 0A\bar{u}$, and both $0\bar{u}$ and $0A\bar{u}$ are the zero vector.
- **Lines stay lines**, and parallel lines stay parallel.
- **Even spacing stays even:** a square grid becomes a grid of identical parallelograms, never curved or bunched.

To see the last two, take points spaced evenly along a line, each one a step $\bar{d}$ further on than the last:

$$\bar{p}, \quad \bar{p} + \bar{d}, \quad \bar{p} + 2\bar{d}, \quad \bar{p} + 3\bar{d}, \quad \dots$$

By the two rules, A sends them to

$$A\bar{p}, \quad A\bar{p} + A\bar{d}, \quad A\bar{p} + 2A\bar{d}, \quad A\bar{p} + 3A\bar{d}, \quad \dots$$

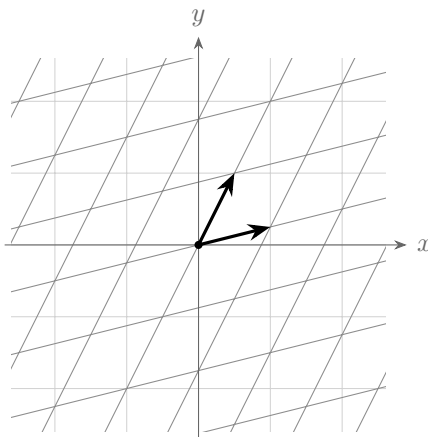
which are again points spaced evenly along a line, now with step $A\bar{d}$. Two parallel lines run in the same direction $\bar{d}$, so their images both run in the direction $A\bar{d}$ and are still parallel. There is one exception, and the formula shows it: if $A\bar{d} = \bar{0}$, every point of the line lands on the single point $A\bar{p}$. That happens only for a matrix that flattens the plane, like the projection in Section 4; there some lines collapse to points and the grid of parallelograms is squashed flat.

Worked example: evenly spaced points stay evenly spaced. Start at $\bar{p} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and step by $\bar{d} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. That gives the points $(0, 1)$, $(1, 2)$, $(2, 3)$, $(3, 4)$, evenly spaced along the line $y = x + 1$. With $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$:

point	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$
image	$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 4 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 7 \\ 9 \end{bmatrix}$	$\begin{bmatrix} 10 \\ 12 \end{bmatrix}$

Each image is the previous one plus $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ is exactly $A\bar{d} = 1\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, as the general argument predicted. The images all satisfy $y = x + 2$ (check: $3 = 1 + 2$, $6 = 4 + 2$, $9 = 7 + 2$, $12 = 10 + 2$), so they lie on a line, evenly spaced with the bigger step $A\bar{d}$. The line was moved and stretched, but it is still a line.

The picture below shows all three consequences at once, for the matrix $\begin{bmatrix} 1 & 0.5 \\ 0.25 & 1 \end{bmatrix}$. The light square grid is carried to the dark grid: every grid line is still a straight line, lines that were parallel still are, every cell became the same parallelogram, and the origin (the dot) did not move. The two arrows are the images of $\hat{i}$ and $\hat{j}$, the sides of every dark cell; Section 3 explains why they matter so much.



The square grid (light) and its image under $\begin{bmatrix} 1 & 0.5 \\ 0.25 & 1 \end{bmatrix}$ (dark).

Watch out

“Linear” here is stricter than “linear function” in Unit 1. There, a linear function was $f(x) = mx + b$, anything whose graph is a straight line. As a transformation, f is linear only when $b = 0$, because the origin must stay put: $f(0) = b$ has to be 0.

Converting a temperature from Celsius to Fahrenheit, $F = 1.8C + 32$, has a straight-line graph but is **not** a linear transformation. It moves the origin (0°C is 32°F), and it breaks the scaling rule: 10°C is 50°F , but 20°C is 68°F , not 100°F . Converting a weight from kilograms to pounds,

$w_{\text{lb}} = 2.2046 w_{\text{kg}}$, has no added constant, and it **is** linear: double the kilograms and you double the pounds.

Non-examples. Here are rules that are *not* linear transformations, and exactly where each one fails.

1. **Translation.** A **translation** slides the whole plane sideways, so every point moves by the same amount, including the origin. For example, $T(x, y) = (x + 1, y + 2)$ slides everything 1 right and 2 up, and $T(0, 0) = (1, 2)$. But every matrix leaves the origin where it is, so **no matrix** can do a translation. It breaks the sum rule too: T applied to $(1, 0) + (0, 1) = (1, 1)$ gives $(2, 3)$, while $T(1, 0) + T(0, 1) = (2, 2) + (1, 3) = (3, 5)$, because adding two moved points counts the slide twice. Graphics systems fake a translation with a third coordinate, a trick beyond this unit. This is also why the scan problem measured pixels from the point being turned about: turning about any other point would also slide the origin.
2. **Squaring a coordinate.** $Q(x, y) = (x^2, y)$ does keep the origin, since $Q(0, 0) = (0, 0)$, but it breaks the scaling rule. Doubling first gives $Q(2, 0) = (4, 0)$; moving first and then doubling gives $2Q(1, 0) = (2, 0)$. Keeping the origin fixed is **necessary** for linearity, but it is not enough.
3. **The temperature conversion** in the box above: a straight-line graph that misses the origin.

Notebook U2-4_Transformations-3_UnitCircleAnimation.ipynb, §3, four deformed grids

2.3 The Same Product Also Changes Coordinates

In Unit 2-2 we found the coordinates of the point $(3, 3)$ in the basis

$$\bar{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \bar{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

that is, the amounts c_1, c_2 with $c_1\bar{b}_1 + c_2\bar{b}_2 = \bar{p}$, where $\bar{p}$ is the vector to $(3, 3)$. The answer was $c_1 = 2$ and $c_2 = 1$. Unit 2-3 then wrote the same problem as one product, $B\bar{c} = \bar{p}$, where B holds the basis vectors as its columns and $\bar{c}$ holds the amounts. Read that product as a transformation:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}}_{B: \text{ the basis as columns}} \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\bar{b} \text{ coordinates}} = \underbrace{\begin{bmatrix} 3 \\ 3 \end{bmatrix}}_{\text{standard coordinates}}$$

The same product tells two different stories.

- **Moving a point:** B sends the point $(2, 1)$ to the point $(3, 3)$. This is Reading 2 from above.
- **Relabelling a point:** the point never moved. $(2, 1)$ is its address in the $\bar{b}$ basis (“go 2 of $\bar{b}_1$ and 1 of $\bar{b}_2$ ”), and $(3, 3)$ is its address in the standard basis (“go 3 right and 3 up”). B **translates between the two languages**.

Both readings are correct; the problem decides which one you use. PCA uses the second reading: it finds a better basis for the data and rewrites every point in it. Going the other way, from standard coordinates back to $\bar{b}$ coordinates, needs the **inverse** of B , which is the subject of Unit 2-5. For

a single point, though, you can already go back: that is just the system Unit 2-2 solved, as the second example below shows.

Worked example: another address in the same basis. Which point has $\bar{b}$ coordinates $(1, 2)$? Multiply by B :

$$B \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Walk it to see why: one step along $\bar{b}_1$ lands at $(2, 1)$, and two steps along $\bar{b}_2$ add $(-2, 2)$, ending at $(0, 3)$. So the point whose standard address is $(0, 3)$ has the address $(1, 2)$ in the $\bar{b}$ basis.

Worked example: going back by solving a system. Now run it backwards: given the standard address $(0, 3)$, find the $\bar{b}$ address (c_1, c_2) . We need $c_1\bar{b}_1 + c_2\bar{b}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, which, entry by entry, is the system

$$\begin{aligned} 2c_1 - c_2 &= 0 \\ c_1 + c_2 &= 3. \end{aligned}$$

Adding the two equations eliminates c_2 : $3c_1 = 3$, so $c_1 = 1$, and then $c_2 = 3 - c_1 = 2$. The address is $(1, 2)$, matching the previous example. Going forward took one product; going back took a small system. Unit 2-5 will turn going back into one product as well.

Worked example: a basis that answers a clinical question. Nurses often measure systolic blood pressure on both arms, so each patient gives a point (L, R) in mmHg. A useful basis separates what the two arms share from how they differ:

$$\bar{s} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ (both arms together),} \quad \bar{d} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ (left up, right down).}$$

In this basis a patient's coordinates (a, b) mean “a common level of a , with a split of b each way.” The matrix with these columns turns them into the two readings. A patient with common level 130 and split 5 has

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 130 \\ 5 \end{bmatrix} = 130 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 135 \\ 125 \end{bmatrix}.$$

135 on the left arm and 125 on the right. Going back, a patient with readings $(142, 138)$ needs $a + b = 142$ and $a - b = 138$; adding gives $2a = 280$, so $a = 140$ and $b = 2$. In the new language this patient has a common level of 140 and arms that differ by $2b = 4$ mmHg, which is the number a clinician checking for an inter-arm difference actually wants. The patient is the same; only the description changed. PCA does the same kind of thing, except that it chooses the basis from the data itself.

Watch out

B turns $\bar{b}$ coordinates into standard coordinates, not the other way round. Feeding B the standard address of our point, $(3, 3)$, gives $B \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$, which is not the $\bar{b}$ address $(2, 1)$ and has nothing to do with it. Before multiplying, ask which language the vector in your hand is written in.

2.4 More Practice: Matrices as Functions

Exercise 2.1

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$.

1. Compute $A \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $A \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $A \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, the last one with both the row reading and the column reading.
2. Find the image of the unit square, whose corners are $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, by multiplying once.

Solution to Exercise 2.1

1. Multiplying by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ takes 1 of the first column and 0 of the second, so $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, the first column. Likewise $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, the second column. For $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$, the column reading gives

$$A \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} - 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 - 2 \\ -3 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix},$$

and the row reading gives $1 \cdot 3 + 2 \cdot (-1) = 1$ and $(-1) \cdot 3 + 1 \cdot (-1) = -4$. They agree: the point $(3, -1)$ is moved to $(1, -4)$.

2. Put the four corners in the columns of X :

$$AX = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 3 & 2 \\ 0 & -1 & 0 & 1 \end{bmatrix}.$$

The square goes to the four-sided shape with corners $(0, 0)$, $(1, -1)$, $(3, 0)$, $(2, 1)$. Its sides are the images of the square's sides, and since parallel lines stay parallel, it is a parallelogram. Notice the origin stayed put, as it must.

Exercise 2.2

Which of these rules are linear transformations? For each one that is, write its matrix; for each one that is not, show a rule it breaks.

$$(a) T(x, y) = (3x, x - y) \quad (b) T(x, y) = (x + y, 1) \quad (c) T(x, y) = (y, x^2)$$

$$(d) T(x, y) = (0, 0) \quad (e) T(x, y) = (2y, -x)$$

Solution to Exercise 2.2

A rule is a matrix transformation when each output entry is a fixed combination of x and y with no added constants and no powers or products of the inputs. Then the row reading hands us the matrix: the coefficients of the first output entry form row 1, those of the second form row 2.

(a) **Linear.** $3x = 3x + 0y$ and $x - y = 1x + (-1)y$, so the matrix is $\begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$.

(b) **Not linear.** The constant 1 moves the origin: $T(0, 0) = (0, 1)$.

(c) **Not linear.** The square breaks the scaling rule: $T(2, 0) = (0, 4)$, but $2T(1, 0) = 2(0, 1) =$

$(0, 2)$.

(d) **Linear.** It is the zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, which sends every vector to $\bar{0}$. Crushing everything to the origin is a (very destructive) linear transformation.

(e) **Linear.** $2y = 0x + 2y$ and $-x = -1x + 0y$, so the matrix is $\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix}$.

Rule (d) is a useful reminder that “linear” does not mean “interesting”: the check is only whether the two rules hold, and for the zero map both sides of each rule are always $\bar{0}$.

Exercise 2.3

Use the tablets-to-doses matrix $D = \begin{bmatrix} 10 & 20 & 0 \\ 5 & 0 & 25 \end{bmatrix}$ from Section 2.

1. What daily dose does the regimen $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ (one of tablet 1, two of tablet 2) give?
2. A caregiver keeps the tablets for two patients in one box: one patient takes $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ a day and the other $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Find the total dose in a day's box in two ways: by adding the regimens first, and by adding the two patients' doses.
3. Without computing anything, what dose does the regimen $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ give, and which property of linear transformations says so?

Solution to Exercise 2.3

1. Take 1 of the first column and 2 of the second: $D \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} 10 \\ 5 \end{bmatrix} + 2 \begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} 50 \\ 5 \end{bmatrix}$, so 50 mg of drug A and 5 mg of drug B.
2. Adding the regimens first, the box holds $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$, and

$$D \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = 3 \begin{bmatrix} 10 \\ 5 \end{bmatrix} + 1 \begin{bmatrix} 20 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 25 \end{bmatrix} = \begin{bmatrix} 50 \\ 65 \end{bmatrix}.$$

Adding the doses instead: the first patient's dose is $\begin{bmatrix} 40 \\ 35 \end{bmatrix}$ (computed in the notes), and the second's is

$$D \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \end{bmatrix},$$

for a total of $\begin{bmatrix} 50 \\ 65 \end{bmatrix}$. The same, as the sum rule guarantees: $D(\bar{u} + \bar{v}) = D\bar{u} + D\bar{v}$.

3. No tablets means no drug: the dose is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. That is the consequence $A\bar{0} = \bar{0}$: a linear transformation always sends the zero vector to the zero vector.

Exercise 2.4

Use the basis $\bar{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and its matrix $B = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ from Unit 2-2.

1. Which point has $\bar{b}$ coordinates $(3, -1)$?

2. What are the $\bar{b}$ coordinates of the point $(1, 2)$?

Solution to Exercise 2.4

1. Going from $\bar{b}$ coordinates to standard ones is one product with B :

$$B \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6+1 \\ 3-1 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

Three steps along $\bar{b}_1$ and one step backwards along $\bar{b}_2$ reach the point $(7, 2)$.

2. Going the other way means solving $c_1\bar{b}_1 + c_2\bar{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, that is, $2c_1 - c_2 = 1$ and $c_1 + c_2 = 2$. Adding the equations gives $3c_1 = 3$, so $c_1 = 1$ and $c_2 = 2 - 1 = 1$. Check: $\bar{b}_1 + \bar{b}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. So $(1, 2)$ has $\bar{b}$ coordinates $(1, 1)$: one of each basis vector.

3 Reading the Columns

3.1 Where Do $\hat{i}$ and $\hat{j}$ Land?

Recall the two standard unit vectors, one unit along each axis:

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Feed a general 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ each unit vector in turn, and use the column reading:

$$A\hat{i} = 1 \begin{bmatrix} a \\ c \end{bmatrix} + 0 \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}, \quad A\hat{j} = 0 \begin{bmatrix} a \\ c \end{bmatrix} + 1 \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

In each product one column is multiplied by 1 and the other by 0, so the product simply picks out a column. Recall that the output of a transformation is called the **image** of the input: the image of a vector is where it lands.

The key fact of this lecture

The columns of A are the places $\hat{i}$ and $\hat{j}$ are sent. Column 1 is the image of $\hat{i}$; column 2 is the image of $\hat{j}$.

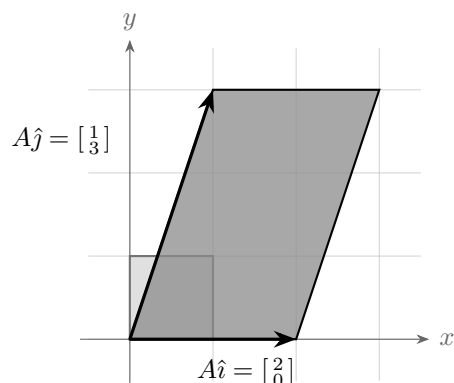
Why does that tell you everything? Every vector is a combination of $\hat{i}$ and $\hat{j}$, since $\begin{bmatrix} x \\ y \end{bmatrix} = x\hat{i} + y\hat{j}$. And A respects combinations (the two rules of linearity), so

$$A(x\hat{i} + y\hat{j}) = x(A\hat{i}) + y(A\hat{j}).$$

Know where the two unit arrows go and you know where **everything** goes: the point (x, y) lands at x of the first column plus y of the second.

In plain words, a 2×2 matrix is a recipe card with two lines on it: “send $\hat{i}$ here” and “send $\hat{j}$ there.” Every other point follows the recipe automatically, because every other point is built out of $\hat{i}$ and $\hat{j}$.

The picture shows this for the matrix from Section 2, $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. The unit square has sides $\hat{i}$ and $\hat{j}$; its image is the parallelogram whose sides are $A\hat{i}$ and $A\hat{j}$, the two columns.



$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$: the unit square (light) goes to a parallelogram (dark).

Notebook U2-4_Transformations-1_MatrixAsMap.ipynb, §2, where the unit vectors land

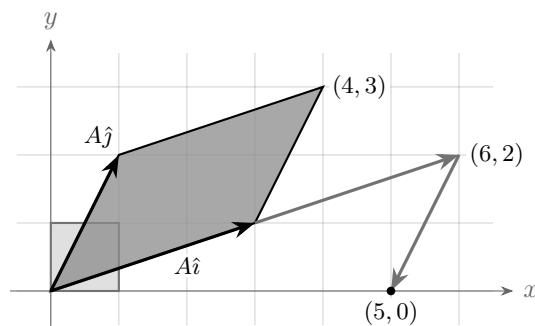
Worked example: reading a new matrix. Read $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ column by column. The first column says $\hat{i}$ lands at $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$; the second says $\hat{j}$ lands at $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The unit square, with corners $\bar{0}$, $\hat{i}$, $\hat{i} + \hat{j}$, $\hat{j}$, therefore goes to the parallelogram with corners

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

where the third corner is the image of $\hat{i} + \hat{j}$, which by the sum rule is $A\hat{i} + A\hat{j}$. Where does the point $(2, -1)$ go? Take 2 of the first column and -1 of the second:

$$A \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}.$$

In words: two steps along the new $\hat{i}$ and one step backwards along the new $\hat{j}$ happen to land on the x -axis, at $(5, 0)$.



$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$: the unit square (light) and its image (dark). The gray walk takes two steps along $A\hat{i}$, to $(6, 2)$, then one step backwards along $A\hat{j}$, ending at $(5, 0)$.

Walking the new grid. The column reading has a picture of its own. Transform the whole square grid with a matrix: each grid line becomes a line of a new, slanted grid whose cells are all copies

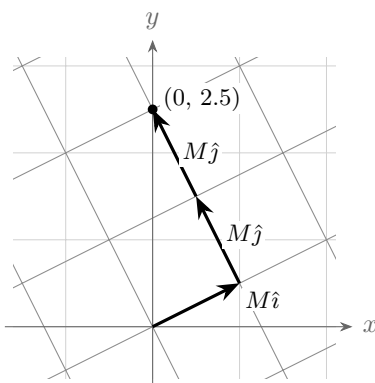
of the parallelogram spanned by the two columns. To find where a point (x, y) lands, start at the origin and walk x cells along the first new direction, then y cells along the second. The figure does this for

$$M = \begin{bmatrix} 1 & -0.5 \\ 0.5 & 1 \end{bmatrix}$$

and the point $(1, 2)$: one step along $M\hat{i} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$, then two steps along $M\hat{j} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$, which ends at

$$M \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 2 \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2.5 \end{bmatrix}.$$

The walk on the new grid is the same as the walk to $(1, 2)$ on the old grid (one step right, two steps up), carried over by the matrix. That is why a picture of just two landed arrows is a complete description of the transformation.



The square grid (light) and its image under M (dark). One step along $M\hat{i}$, then two along $M\hat{j}$, reach $M \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Watch out

The images are the columns, not the rows. For $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ the first *row* is $(2, 1)$, but $A\hat{i} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ is the first *column*. Reading the rows as the images of $\hat{i}$ and $\hat{j}$ gives you the transpose $A^\top$, a different transformation (here $A^\top$ would send $\hat{i}$ to $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$). The two readings agree only for a symmetric matrix ($A^\top = A$, Unit 2-3), such as $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ above, which is exactly why it is dangerous to practice only on symmetric examples.

3.2 Going Backwards: Build the Matrix You Want

The key fact also runs backwards. Designing a transformation comes down to two decisions: where should $\hat{i}$ go, and where should $\hat{j}$ go? Write the answers as the columns of a matrix and you are finished.

Worked example: a quarter turn counterclockwise. $\hat{i}$ points right along the x -axis; turning it a quarter turn counterclockwise points it straight up. $\hat{j}$ points up; turning it a quarter turn points it left. So

$$\hat{i} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \implies \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

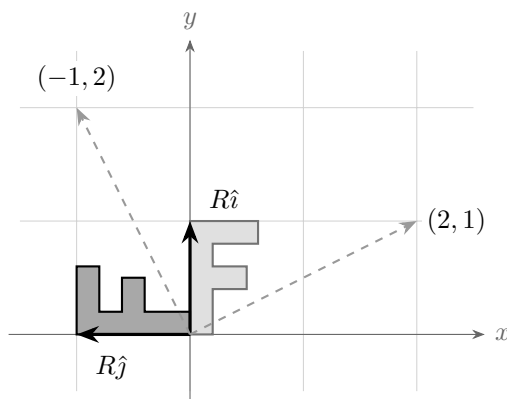
where the arrow $\mapsto$ is read “is sent to.”

Check on a point. By the column reading,

$$R \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix},$$

which is the point $(2, 1)$ turned a quarter turn, as the picture confirms.

The picture, like most in these notes, uses a deliberately lopsided letter F as a test shape. It has no symmetry, so if a transformation flips it or turns it, you cannot miss it.



The F before (light) and after R (dark); the columns of R drawn as arrows.

No formula was looked up here. Every matrix in the catalogue that follows was built the same way.

Notebook U2-4_Transformations-1_MatrixAsMap.ipynb, §4, a quarter turn built on purpose

Here are four more constructions, from easy to harder. Each one is the same two decisions.

Worked example: stretch sideways by 3. Widths should triple and heights should not change. So $\hat{i}$ should become three times as long in the same direction, and $\hat{j}$ should stay where it is:

$$\hat{i} \mapsto \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \Longrightarrow \quad \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Check on $(2, 5)$: $2 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$. The x -coordinate tripled and the y -coordinate is untouched, just what happens to an image stretched to fill a wider screen.

Worked example: a quarter turn clockwise. Turning clockwise, $\hat{i}$ (pointing right) ends up pointing straight down, and $\hat{j}$ (pointing up) ends up pointing right:

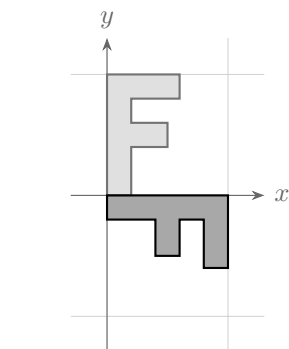
$$\hat{i} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \Longrightarrow \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Check on $(2, 1)$: $2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. The point $(2, 1)$, a little above the x -axis, turns clockwise to $(1, -2)$, below it. Compare with the counterclockwise $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$: the two matrices differ only in the signs of the off-diagonal entries.

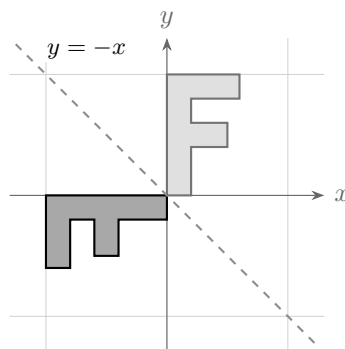
Worked example: a flip across the line $y = -x$. The mirror is the line through the origin that runs down to the right at 45° . The mirror image of $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ across it is $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$, and the mirror image of $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ (picture each one folding over the dashed line in the figure). So

$$\hat{i} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \Longrightarrow \quad \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

Check on $(3, 1)$: $3 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$. In general the matrix sends (x, y) to $(-y, -x)$, and a point on the mirror, such as $(1, -1)$, goes to $(1, -1)$: it stays put, as a point on a mirror should. We will meet this matrix again in Section 6.



Quarter turn clockwise $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$



Flip across $y = -x$: $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Worked example (harder): when you are told where other vectors go. Sometimes the description does not mention $\hat{i}$ and $\hat{j}$ at all. Suppose a transformation is known to send

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

To use the column recipe we need the images of $\hat{i}$ and $\hat{j}$, so write the unit vectors as combinations of the two vectors we know about:

$$\hat{i} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \hat{j} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Linearity says the images combine in the same way:

$$A\hat{i} = \frac{1}{2} \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad A\hat{j} = \frac{1}{2} \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

so

$$A = \begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}.$$

Check both facts we started from: $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 2-1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2-1 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. The lesson: knowing where any two independent vectors (Unit 2-2) land pins down the whole transformation, because every vector, $\hat{i}$ and $\hat{j}$ included, is a combination of them. The unit vectors are just the most convenient pair to know about.

Watch out

Images go in as columns, and in order. Two slips turn a correct design into a different transformation.

- **Writing the images as rows.** For the counterclockwise quarter turn, putting $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ in as rows gives $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the *clockwise* quarter turn: exactly the opposite rotation.
- **Swapping the columns.** Putting $\hat{j}$'s image first gives $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, which is not a turn at all

but the flip across the y -axis.

A ten-second habit catches both: multiply your matrix by $\hat{i}$ and make sure the answer is where you wanted $\hat{i}$ to go.

Now practice reading matrices and building them.

Exercise 3.1

1. For each matrix, say where $\hat{i}$ and $\hat{j}$ land and describe the transformation in words:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

2. Write the matrix that sends $\hat{i}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\hat{j}$ to $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Where does it send $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$? And $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$?
3. A matrix sends every point to a point on the x -axis. What must its second row be?

Solution to Exercise 3.1

1. Read each matrix column by column.

- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$: $\hat{i}$ stays at $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{j}$ goes to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$. Horizontal directions are untouched and vertical ones are reversed, so the point (x, y) goes to $(x, -y)$: the plane is **flipped across the x -axis**.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: $\hat{i} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The two unit vectors trade places, so (x, y) goes to (y, x) : the two coordinates are **swapped**, which is a flip across the line $y = x$.
- $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$: $\hat{i} \mapsto \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Both arrows are tripled in place, so (x, y) goes to $(3x, 3y)$: everything is **scaled by 3**, with directions unchanged.

2. Stack the two images as columns:

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

The vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is $\hat{i}$, so it goes to the first column, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. For $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, take 2 of the first column plus 3 of the second:

$$A \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 3 \\ 2 + 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$

Geometrically, this matrix turns by 45° and stretches by $\sqrt{2}$: both columns have length $\sqrt{1^2 + 1^2} = \sqrt{2}$, and each sits 45° counterclockwise from where it started ($\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is 45° above the x -axis, and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is 45° past the y -axis).

3. Write the matrix as $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$. By the column reading,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix},$$

so the second entry of every output is $cx + dy$. A point is on the x -axis exactly when its second coordinate is 0, so we need $cx + dy = 0$ for all x and y . Taking $(x, y) = (1, 0)$ gives $c = 0$, and taking $(x, y) = (0, 1)$ gives $d = 0$. So $c = d = 0$: the **second row is all zeros**.

The key fact explains why. Both columns (the images of $\hat{i}$ and $\hat{j}$) then lie on the x -axis, and every output is a combination of the columns, so it must land on the x -axis too.

3.3 More Practice: Reading and Building

Exercise 3.2

Let $G = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$. Where do $\hat{i}$ and $\hat{j}$ land? Where does $(1, 2)$ go? What shape does the unit square become?

Solution to Exercise 3.2

Read the columns: $\hat{i} \mapsto \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, so $\hat{j}$ does not move at all. The point $(1, 2)$ goes to

$$G \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The unit square goes to the parallelogram with corners $\bar{0}$, $G\hat{i} = (2, 1)$, $G\hat{i} + G\hat{j} = (2, 2)$, and $G\hat{j} = (0, 1)$. Its left side is still the upright segment from $(0, 0)$ to $(0, 1)$, while its bottom side has been stretched to $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, of length $\sqrt{5}$, and tilted upward. So G stretches horizontally and slides points up in proportion to how far right they are.

Exercise 3.3

Write the matrix for each transformation by deciding where $\hat{i}$ and $\hat{j}$ go.

1. Shrink everything to half its size.
2. Flip across the y -axis and double every height.
3. Move every point straight sideways onto the y -axis (keep its height, set its x -coordinate to 0).

Solution to Exercise 3.3

1. Both unit vectors keep their directions and halve their lengths: $\hat{i} \mapsto \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$, $\hat{j} \mapsto \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$, so the matrix is $\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} = 0.5I$.
2. The flip reverses $\hat{i}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$; doubling heights sends $\hat{j}$ to $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$. The matrix is $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, and it sends (x, y) to $(-x, 2y)$.
3. Every point keeps its height and loses its x -coordinate. So $\hat{i} = (1, 0)$, which has height 0, goes to the origin, and $\hat{j}$ stays put: $\hat{i} \mapsto \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\hat{j} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The matrix is $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, which sends (x, y) to $(0, y)$. (This is a projection, a member of the catalogue in Section 4.)

Exercise 3.4

A transformation sends $(1, 0)$ to $(4, 1)$ and $(1, 1)$ to $(5, 3)$. Find its matrix, and the image of $(2, -1)$.

Solution to Exercise 3.4

The first fact gives the first column directly: $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 4 \\ 1 \end{bmatrix}$. For the second column, write $\hat{j}$ as a combination of the two vectors we know about: $\hat{j} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. By linearity its image is the same combination of their images:

$$A\hat{j} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} - \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \text{so} \quad A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}.$$

Check: $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+1 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Then $A \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 4 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}$, so $(2, -1)$ lands at $(7, 0)$ on the x -axis.

Exercise 3.5

A matrix leaves every point of the x -axis exactly where it is. What must its first column be? What if it leaves *every* point of the plane where it is?

Solution to Exercise 3.5

The point $(1, 0)$ is on the x -axis, so $A\hat{i} = \hat{i}$: the first column must be $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. That is also enough for the whole axis, since any point $(x, 0)$ is $x\hat{i}$ and goes to $x A\hat{i} = (x, 0)$. The second column is free, so matrices like $\begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix}$ also fix the x -axis (they move everything else).

If every point stays put, then in particular $\hat{j}$ does too, so the second column is $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and the matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$. The identity is the only matrix that moves nothing.

4 The Catalogue

4.1 Scaling and Shear

Every matrix in this section is built the way Section 3 taught: decide where $\hat{i}$ and $\hat{j}$ go, and stack the answers as columns.

Scaling: a diagonal matrix. The matrix

$$\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

sends $\hat{i}$ to $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ and $\hat{j}$ to $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$: $\hat{i}$ doubles and $\hat{j}$ halves. Each axis is stretched by its own factor, so the F comes out twice as wide and half as tall. A matrix like this, with nonzero entries only on the diagonal, is a **diagonal** matrix (Unit 2-3). When the two factors are equal, $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = kI$, every direction is stretched by the same amount and the shape is enlarged uniformly. A negative factor flips that axis over.

Shear: slide the top sideways. The matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

leaves $\hat{i}$ where it is and tips $\hat{j}$ over to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. It sends $\begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} x+y \\ y \end{bmatrix}$: a point's height y does not change, but the point slides horizontally by an amount proportional to its height (here, equal to it). Points on the x -axis do not move at all, and the higher a point, the further it slides, like a deck of cards pushed over.



Light F: the original. Dark F: its image. Check each matrix's columns against where the F's bottom edge (which runs along $\hat{i}$) and left edge (which runs along $\hat{j}$) went.

Notebook U2-4_Transformations-2_NamedTransforms.ipynb, §1, the full panel of named transforms

Scaling in general. The diagonal matrix

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

multiplies the first coordinate by s_x and the second by s_y , and does nothing else. Its columns say the same thing: $\hat{i}$ stays on the x -axis and $\hat{j}$ stays on the y -axis, each with a new length (and, if its factor is negative, a new direction).

Worked example: uniform enlargement. $1.5I = \begin{bmatrix} 1.5 & 0 \\ 0 & 1.5 \end{bmatrix}$ sends $(2, -4)$ to $(3, -6)$. The image points in the same direction as the original, one and a half times as far out; applied to a picture, this is zooming in by 50% about the origin.

Worked example: changing units is scaling. A cohort's data table records each patient's weight in kilograms and height in centimeters, and a partner hospital wants pounds and inches. Since 1 kg is about 2.2046 lb and 1 cm is about 0.3937 in, the conversion is the diagonal matrix

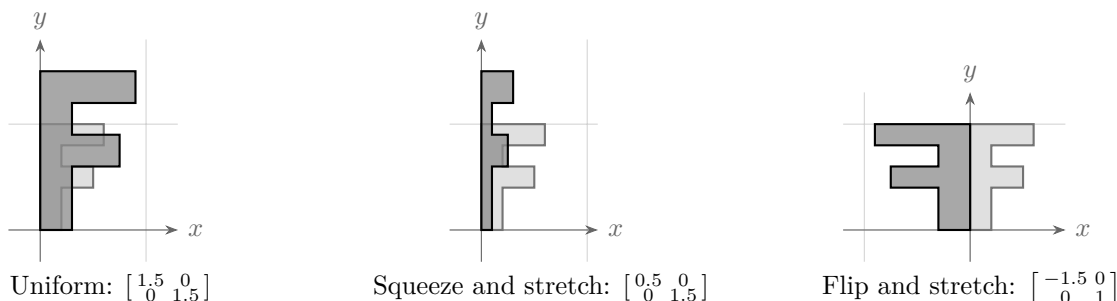
$$\begin{bmatrix} 2.2046 & 0 \\ 0 & 0.3937 \end{bmatrix} \begin{bmatrix} 70 \\ 170 \end{bmatrix} = \begin{bmatrix} 2.2046 \times 70 \\ 0.3937 \times 170 \end{bmatrix} \approx \begin{bmatrix} 154.3 \\ 66.9 \end{bmatrix}$$

for a patient who weighs 70 kg and is 170 cm tall: about 154.3 lb and 66.9 in. Every change of units in a data table is a diagonal matrix, with one factor per column of the table, and converting the whole table at once is one matrix product.

Worked example: a negative factor. $\begin{bmatrix} -1.5 & 0 \\ 0 & 1 \end{bmatrix}$ sends $(2, 3)$ to $(-3, 3)$: the x -coordinate is multiplied by -1.5 , so the point jumps to the other side of the y -axis and ends up half again as far

from it, while its height is unchanged. A negative scaling factor is a flip and a stretch in one move, as the third panel below shows: the F is mirrored and widened.

A zero factor. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ multiplies every height by 0, crushing the plane onto the x -axis. That is the projection of Section 4.3: the entries of the catalogue overlap, and that is fine.



Watch out

Standardizing a variable is not a matrix. The z -score from Unit 1-3, $z = (x - \text{mean})/\text{sd}$, looks like a rescaling, but it is two steps. Dividing by the standard deviation is a scaling, a diagonal matrix. Subtracting the mean is a translation, and no matrix can do that. This is why data pipelines **center** the data first, as a separate step, and why PCA begins by centering: once the data are centered, everything that follows is linear.

Shear in general. The horizontal shear with factor k ,

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ky \\ y \end{bmatrix},$$

slides each point sideways by k times its height: to the right when $k > 0$, to the left when $k < 0$. The **vertical shear** swaps the roles of the axes:

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y + kx \end{bmatrix},$$

so each point slides *up* by k times its x -coordinate. Read it from the columns: $\hat{i}$ tips up to $\begin{bmatrix} 1 \\ k \end{bmatrix}$, while $\hat{j}$ stays put, so the y -axis does not move.

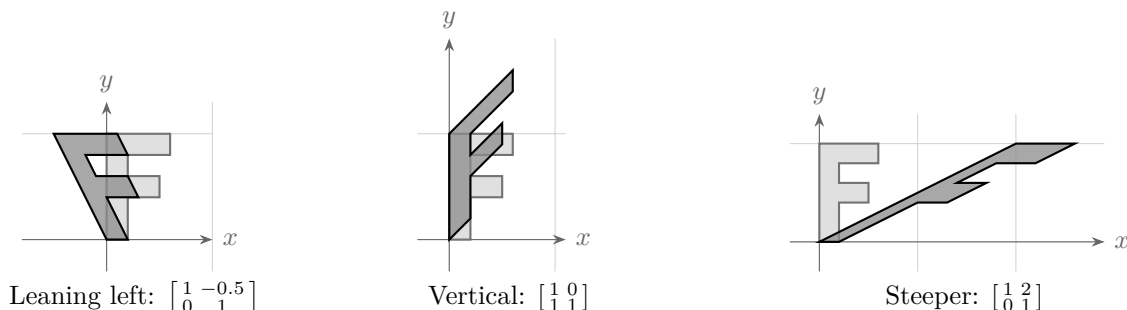
Worked example: a steeper shear. With $k = 2$, the shear $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ sends

$$(1, 1) \mapsto (1 + 2 \cdot 1, 1) = (3, 1), \quad (0, 2) \mapsto (0 + 2 \cdot 2, 2) = (4, 2), \quad (3, 0) \mapsto (3, 0).$$

The point at height 2 slides 4 units, the point at height 1 slides 2, and the point on the x -axis does not move. The slide is proportional to height, which is exactly what makes a shear a shear.

Worked example: leaning left. With $k = -0.5$, the shear $\begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix}$ sends $(0, 1)$ to $(-0.5, 1)$ and $(2, 2)$ to $(2 - 1, 2) = (1, 2)$. Each point slides left by half its height, so the F leans back instead of forward (first panel below).

Worked example: a vertical shear. With $k = 1$, the vertical shear $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ sends $(1, 0)$ to $(1, 1)$ and $(2, 1)$ to $(2, 1 + 2) = (2, 3)$. Each point rises by its own x -coordinate, so points on the y -axis stay put and points further right climb further (second panel).



Shears are more common than they look. Italic type is a shear of upright type. A camera that reads its image one row at a time while the subject moves sideways records a slanted, sheared picture (the “rolling shutter” effect). And image libraries offer shear as a standard augmentation, alongside turns and flips.

Watch out

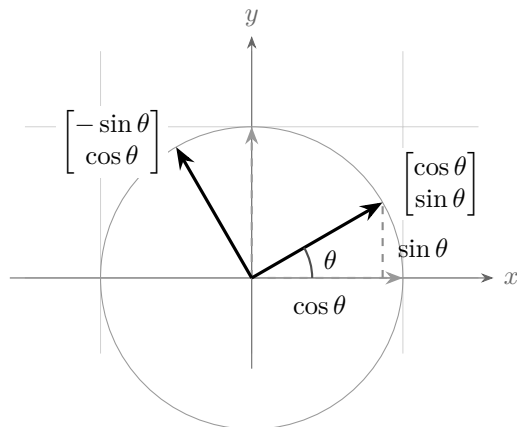
A shear is not a rotation, even though the F looks tilted. Under $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ the F’s horizontal bars stay horizontal and keep their lengths (every point of a bar slides by the same amount), while its upright stem tips over and gets longer: it ran from $(0,0)$ to $(0,1)$ and now runs from $(0,0)$ to $(1,1)$, of length $\sqrt{2}$. A rotation would tilt the bars too, and would keep every length. Section 5 turns this observation into a test.

4.2 Rotation by Any Angle

The quarter turn was one rotation. To rotate by any angle θ , again ask where the two unit vectors land.

Turn $\hat{i}$ counterclockwise by θ . Turning does not change length, so it is still a unit vector, now at angle θ from the positive x -axis. Recall how cosine and sine are defined: for an arrow of length 1 at angle θ from the positive x -axis, $\cos \theta$ is its horizontal coordinate and $\sin \theta$ its vertical coordinate. So $\hat{i}$ lands at

$$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}.$$



Both unit vectors turned by θ (dashed: where they started); their new coordinates are the columns.

The vector $\hat{j}$ starts a quarter turn ahead of $\hat{i}$, and turning both by the same angle keeps it a quarter turn ahead. So $\hat{j}$ lands where the new $\hat{i}$ does after one more quarter turn. The quarter turn R from Section 3 sends $\begin{bmatrix} a \\ b \end{bmatrix}$ to $a \begin{bmatrix} 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -b \\ a \end{bmatrix}$, so $\hat{j}$ lands at

$$\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}.$$

The rotation matrix

Stack the two images as columns. Rotation counterclockwise by θ is

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

As a check, at $\theta = 90^\circ$ we have $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$, so $R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, the quarter turn we built by hand.

Values worth knowing. A handful of angles have exact cosines and sines, and they are the ones that come up again and again. Unit 1-3 measured angles in radians; the table gives both.

θ (degrees)	0°	30°	45°	60°	90°	180°	270°
θ (radians)	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$
$\cos \theta$	1	$\frac{\sqrt{3}}{2} \approx 0.866$	$\frac{1}{\sqrt{2}} \approx 0.707$	$\frac{1}{2}$	0	-1	0
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}} \approx 0.707$	$\frac{\sqrt{3}}{2} \approx 0.866$	1	0	-1

Plugging these in gives, for example,

$$R(45^\circ) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad R(60^\circ) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}, \quad R(180^\circ) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Worked example: the half turn. $R(180^\circ) = -I$ sends (x, y) to $(-x, -y)$; for instance $(3, 1) \mapsto (-3, -1)$. Every point goes to the opposite side of the origin, the same distance away, which is what turning halfway around does.

Worked example: a point on the axis. Turn $(2, 0)$ by 60° . The point is $2\hat{i}$, so its image is 2 times the first column of $R(60^\circ)$:

$$R(60^\circ) \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 1 \\ 1.732 \end{bmatrix}.$$

The image is still at distance 2 from the origin, since $\sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$, and it now sits at 60° above the x -axis.

Worked example: a general point. Turn $(2, 2)$ by 30° . Take 2 of each column of $R(30^\circ)$:

$$R(30^\circ) \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix} + 2 \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} \sqrt{3} - 1 \\ 1 + \sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 0.732 \\ 2.732 \end{bmatrix}.$$

Check that the length did not change. Using $(a - b)^2 = a^2 - 2ab + b^2$,

$$(\sqrt{3} - 1)^2 + (\sqrt{3} + 1)^2 = (4 - 2\sqrt{3}) + (4 + 2\sqrt{3}) = 8,$$

so the image has length $\sqrt{8}$, exactly the length of $(2, 2)$, since $2^2 + 2^2 = 8$. The original point sat at 45° above the x -axis; its image sits at 75° , as you can confirm with a calculator. Turning adds 30° to the angle and leaves the distance alone.

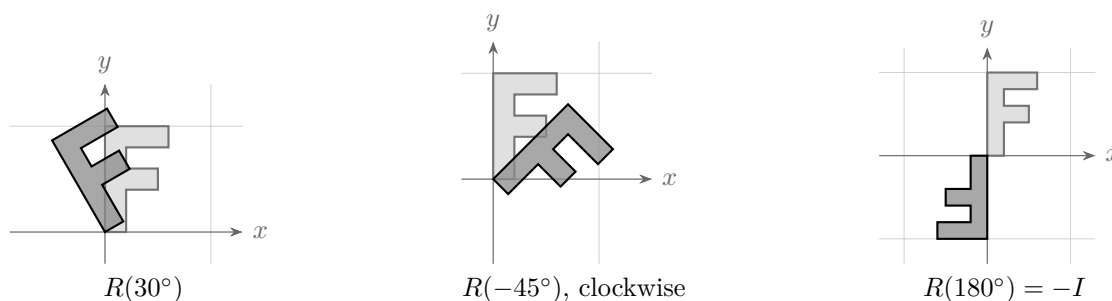
Worked example (harder): turning clockwise. A clockwise turn by θ is a counterclockwise turn by $-\theta$. A point at angle $-\theta$ on the unit circle is the mirror image, below the x -axis, of the point at angle θ , so $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$. Substituting into R ,

$$R(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} :$$

the minus sign moves from the top right to the bottom left. At $\theta = 90^\circ$ this is $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the clockwise quarter turn we built by hand in Section 3.2. As an example, turn $(0, 2)$ clockwise by 30° :

$$R(-30^\circ) \begin{bmatrix} 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} \sin 30^\circ \\ \cos 30^\circ \end{bmatrix} = 2 \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}.$$

The point started on the y -axis, at 90° , and turned clockwise to 60° , still at distance 2. (It lands on the same point as the 60° example above, reached from the other side.)



Watch out

Where the minus sign goes, and what the calculator thinks an angle is.

- For a *counterclockwise* turn the minus sign sits in the **top right**. If you are unsure, check the first column: turning $\hat{i}$ a little counterclockwise lifts it above the x -axis, so the bottom-left entry, $\sin \theta$, must be positive for small positive θ . With the minus sign in the bottom left you have built the clockwise turn instead.
- Python's `np.cos` and `np.sin` expect **radians**. `np.cos(30)` is the cosine of 30 radians, about 0.154, not $\cos 30^\circ \approx 0.866$. Convert first: `np.cos(np.radians(30))`.

The scan problem is solved. “Turn by 30° ” is $R(30^\circ)$, and “enlarge by half” multiplies every length by 1.5, so the matrix we want is $1.5 R(30^\circ)$. With $\cos 30^\circ \approx 0.866$ and $\sin 30^\circ = 0.5$,

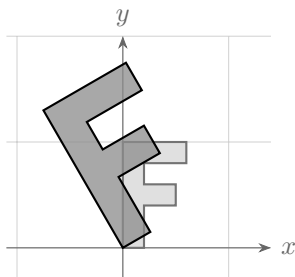
$$A = 1.5 R(30^\circ) = 1.5 \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix} \approx \begin{bmatrix} 1.30 & -0.75 \\ 0.75 & 1.30 \end{bmatrix},$$

and the single product AX moves every pixel of the scan at once.

Now finish the three-pixel scan from Section 1.1. Keeping one more decimal place, $1.5 \cos 30^\circ \approx 1.299$ and $1.5 \sin 30^\circ = 0.75$, so

$$AX \approx \begin{bmatrix} 1.299 & -0.75 \\ 0.75 & 1.299 \end{bmatrix} \begin{bmatrix} 100 & 0 & 100 \\ 0 & 100 & 100 \end{bmatrix} \approx \begin{bmatrix} 129.9 & -75 & 54.9 \\ 75 & 129.9 & 204.9 \end{bmatrix}.$$

Read it column by column: the pixel at $(100, 0)$ moves to about $(129.9, 75)$, the pixel at $(0, 100)$ to about $(-75, 129.9)$, and the pixel at $(100, 100)$ to about $(54.9, 204.9)$. The first two columns are just 100 times the columns of A , as they must be, since those pixels are $100\hat{i}$ and $100\hat{j}$. Check the enlargement on the first pixel: it was 100 from the origin, and now it is $\sqrt{129.9^2 + 75^2} \approx 150$ away, one and a half times as far. The third pixel was $\sqrt{100^2 + 100^2} \approx 141.4$ away and is now about 212.1 away, again 1.5 times as far. Every pixel was turned by the same angle and pushed out by the same factor, which is exactly what the radiologist asked for.



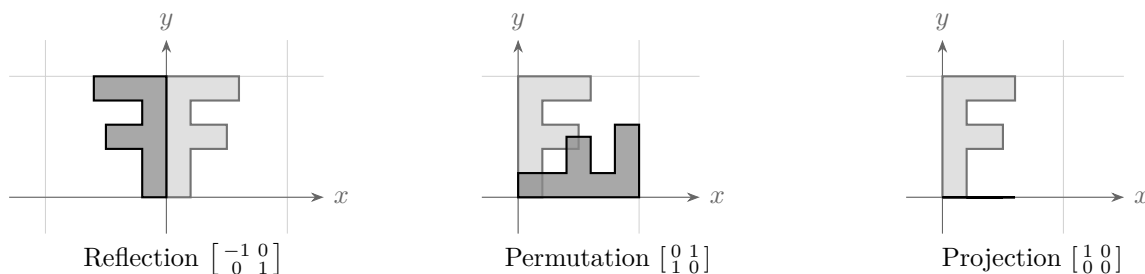
The scan transformation $1.5 R(30^\circ)$ applied to the F: turned by 30° and enlarged by half.

4.3 Reflection, Permutation, and Projection

Reflection. The matrix $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ flips the plane across the y -axis: $\hat{i}$ reverses to $[-1]$ while $\hat{j}$ stays. Each point (x, y) goes to its mirror image $(-x, y)$ on the other side of the y -axis. To flip across the x -axis instead, flip the sign of the other column.

Permutation. The matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ swaps the coordinates: $\hat{i}$ and $\hat{j}$ trade places, and (x, y) goes to (y, x) . A **permutation** is a reordering, and this matrix reorders the two entries of every vector. Geometrically it is a reflection across the line $y = x$.

Projection. The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ keeps $\hat{i}$ and crushes $\hat{j}$ to $\bar{0}$. It sends (x, y) to $(x, 0)$: every point drops straight down (or straight up) onto the x -axis.



Projection destroys information. The whole F lands on a segment of the x -axis, and from that segment there is no way to recover the F: many inputs share each output, so the function cannot

be undone. The identity $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, by contrast, sends $\hat{i}$ to $\hat{i}$ and $\hat{j}$ to $\hat{j}$: it moves nothing at all. Unit 2-5 measures this kind of collapse with the **determinant**, and asks when a transformation can be reversed with an **inverse**.

More reflections. The two axis flips are the simplest mirrors.

- Across the x -axis, $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ sends $(2, 3)$ to $(2, -3)$; across the y -axis, $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ sends $(2, 3)$ to $(-2, 3)$. In image augmentation, the left-right mirror of a photo is the y -axis flip applied to every pixel (with pixel positions measured from the center of the image).
- Across the line $y = -x$, the matrix built in Section 3.2 is $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, which sends (x, y) to $(-y, -x)$. For example, $(2, 5) \mapsto (-5, -2)$.

Worked example (harder): a mirror at 30° . Any line through the origin can serve as a mirror, and its matrix is built the same way. Take the line through the origin at 30° above the x -axis, and reason with angles:

- $\hat{i}$ points along 0° , which is 30° *below* the mirror. Its mirror image is 30° *above* the mirror, at 60° : it lands at $\begin{bmatrix} \cos 60^\circ \\ \sin 60^\circ \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$.
- $\hat{j}$ points along 90° , which is 60° *above* the mirror. Its mirror image is 60° *below* the mirror, at $30^\circ - 60^\circ = -30^\circ$: it lands at $\begin{bmatrix} \cos(-30^\circ) \\ \sin(-30^\circ) \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$.

Stack the images as columns:

$$M_{30} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \approx \begin{bmatrix} 0.5 & 0.866 \\ 0.866 & -0.5 \end{bmatrix}.$$

Two checks confirm it is the right mirror. The point $(\sqrt{3}, 1)$ lies on the mirror (its slope is $1/\sqrt{3} = \tan 30^\circ$), so it should stay put, and it does:

$$M_{30} \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \\ \frac{3}{2} - \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}.$$

The point $(-1, \sqrt{3})$ sits at 120° , perpendicular to the mirror, so it should flip straight through to the opposite side, $(1, -\sqrt{3})$, and it does:

$$M_{30} \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} + \frac{3}{2} \\ -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}.$$

A mirror leaves the points on it alone and reverses the direction straight across it; M_{30} does both.

Mirror or turn? The half turn $R(180^\circ) = -I$ sends (x, y) to $(-x, -y)$, which sounds like a reflection “through the origin.” But look at the $R(180^\circ)$ panel in Section 4.2: the F comes out upside down, not mirror-reversed, and you could read it normally by turning the page around. A true reflection reverses the F, so that it reads backwards however you turn it. So $-I$ belongs with the rotations.

More permutations. In the plane, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ sends $(3, 7)$ to $(7, 3)$. Permutations earn their keep in higher dimensions, where they reorder the features of a data table. Suppose one hospital records

vital signs as (systolic pressure, diastolic pressure, heart rate) and another as (heart rate, systolic, diastolic). The permutation

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad P \begin{bmatrix} 120 \\ 80 \\ 72 \end{bmatrix} = \begin{bmatrix} 72 \\ 120 \\ 80 \end{bmatrix},$$

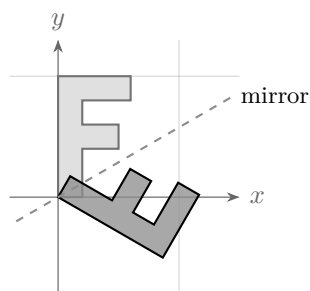
converts one layout to the other. Each row of P has a single 1, marking which old entry to copy into that position: row 1 copies entry 3 (heart rate), row 2 copies entry 1, row 3 copies entry 2. A permutation only reorders; it never changes a value, which is why it keeps every length (Section 5).

More projections. The projection onto the x -axis is an old friend. In Unit 2-2 the wind $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ acting on a plane heading along the x -axis split into a tailwind $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and a crosswind. The tailwind was the shadow of the wind on the heading, and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ computes exactly that shadow, for every vector at once.

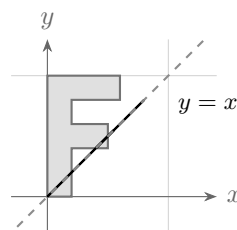
- **Onto the y -axis:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ keeps $\hat{j}$ and crushes $\hat{i}$, so $(4, -2) \mapsto (0, -2)$. In a data table this is “drop the first feature and keep the second.”
- **Onto the line $y = x$ (harder).** Build it from the Unit 2-2 projection formula, $\text{proj}_{\hat{w}}(\bar{v}) = (\bar{v} \cdot \hat{w}) \hat{w}$, with the direction $\hat{w} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ of the line. For $\hat{i}$: $\hat{i} \cdot \hat{w} = \frac{1}{\sqrt{2}}$, so its shadow is $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$. The same computation gives $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ for $\hat{j}$. So the matrix is

$$P_{\text{diag}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix},$$

and it sends (x, y) to $\left(\frac{x+y}{2}, \frac{x+y}{2}\right)$: both coordinates are replaced by their **average**. For example, $(3, 1)$, $(4, 0)$, $(0, 4)$, and $(2, 2)$ all land on $(2, 2)$. Only the average survives; the difference between the coordinates is lost. Applied to the two-arm blood pressures of Section 2, the readings $(142, 138)$ project to $(140, 140)$: the common level is kept and the inter-arm split is thrown away.



Reflection M_{30} across the line at 30°



Projection P_{diag} onto the line $y = x$

Watch out

Projecting is not turning onto the axis. Projection drops each point straight down onto the line, along the perpendicular, and usually shortens it: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ sends $(3, 3)$ to $(3, 0)$, of length 3. Turning $(3, 3)$ down onto the x -axis would keep its length and give $(3\sqrt{2}, 0) \approx (4.24, 0)$ instead. A projection keeps the part of a vector that lies along the line and discards the rest.

The catalogue at a glance. Every entry was built by the same two decisions, so every entry can be read off from its columns.

Move	Matrix	$\hat{i}$ lands at	$\hat{j}$ lands at	What happens to the F
Scaling	$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$	$\begin{bmatrix} s_x \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ s_y \end{bmatrix}$	stretched along each axis
Shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} k \\ 1 \end{bmatrix}$	leans sideways, heights kept
Rotation	$R(\theta)$	$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$	$\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$	turned, same size and shape
Reflection	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	mirrored across the y -axis
Permutation	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	mirrored across $y = x$
Projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	flattened onto a segment
Identity	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	unchanged

Now practice with the catalogue.

Exercise 4.1

1. Rotate the point $(2, 0)$ by 90° counterclockwise using $R(90^\circ)$. Then rotate $(1, 1)$ by 45° using $R(45^\circ)$ (take $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$). Sketch both.
2. Apply the shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ to the four corners of the unit square, $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, and describe the shape you get.
3. Write the matrix that reflects across the x -axis, and check it on $(3, 5)$.
4. Where does the projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ send $(3, 5)$, $(3, -2)$, and $(3, 0)$? What does that tell you about undoing it?

Solution to Exercise 4.1

1. For the quarter turn, use the column reading:

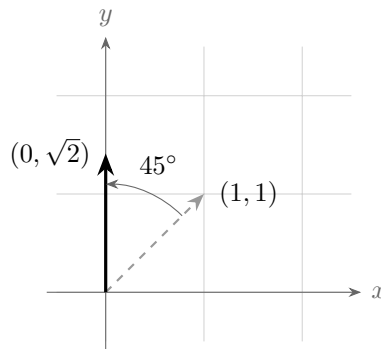
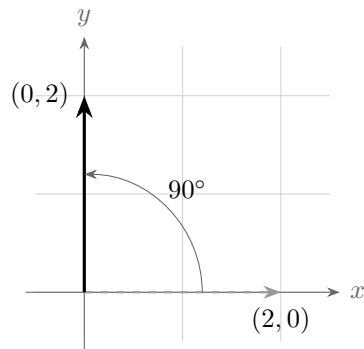
$$R(90^\circ) \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

The point moves from the x -axis straight up to the y -axis, the same distance 2 from the origin.

For the 45° turn, every entry of $R(45^\circ)$ is $\pm \frac{1}{\sqrt{2}}$, so factor that out:

$$R(45^\circ) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}.$$

The diagonal point, which sits at 45° , turns onto the y -axis at 90° , keeping its length $\sqrt{1^2 + 1^2} = \sqrt{2}$.



2. The shear sends (x, y) to $(x + y, y)$:

$$(0, 0) \mapsto (0, 0), \quad (1, 0) \mapsto (1, 0), \quad (1, 1) \mapsto (2, 1), \quad (0, 1) \mapsto (1, 1).$$

The bottom edge (height 0) stays where it is, and the top edge (height 1) slides one unit to the right. The result is a **parallelogram** leaning to the right, with the same base and the same height as the square.

3. Keep $\hat{i}$ and reverse $\hat{j}$, then stack those as columns:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}.$$

The point $(3, 5)$ goes to $(3, -5)$, its mirror image below the x -axis.

4. The projection keeps the first coordinate and replaces the second with 0, so all three points go to $(3, 0)$. Three different inputs share one output, so **no function can send $(3, 0)$ back** to the right one: we cannot tell whether it came from $(3, 5)$, from $(3, -2)$, or from $(3, 0)$ itself. The projection cannot be undone. That is what “destroys information” means, and in Unit 2-5 a zero determinant will detect it.

4.4 More Practice: The Catalogue

Exercise 4.2

1. Rotate $(0, 3)$ counterclockwise by 30° .
2. Rotate $(1, 2)$ by 180° .
3. Rotate $(4, 0)$ *clockwise* by 45° .

Give exact answers, then decimals to three places.

Solution to Exercise 4.2

1. The point is $3\hat{j}$, so its image is 3 times the second column of $R(30^\circ)$:

$$R(30^\circ) \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} -\sin 30^\circ \\ \cos 30^\circ \end{bmatrix} = 3 \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} -3/2 \\ 3\sqrt{3}/2 \end{bmatrix} \approx \begin{bmatrix} -1.5 \\ 2.598 \end{bmatrix}.$$

The point started at 90° and is now at 120° , just past the y -axis, still at distance 3.

2. $R(180^\circ) = -I$, so $(1, 2) \mapsto (-1, -2)$: the point directly opposite through the origin.
 3. Clockwise by 45° is $R(-45^\circ) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$. The point is $4\hat{i}$, so take 4 of the first column:

$$R(-45^\circ) \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \frac{4}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ -2\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 2.828 \\ -2.828 \end{bmatrix},$$

using $\frac{4}{\sqrt{2}} = 2\sqrt{2}$. The point now sits 45° below the x -axis, still at distance 4.

Exercise 4.3

Write the matrix of the vertical shear that lifts every point by half of its x -coordinate. Apply it to the corners of the unit square and describe the result.

Solution to Exercise 4.3

The rule is $(x, y) \mapsto (x, y + 0.5x)$. So $\hat{i} = (1, 0)$ is lifted to $(1, 0.5)$ and $\hat{j} = (0, 1)$, with $x = 0$, does not move: the matrix is $\begin{bmatrix} 1 & 0 \\ 0.5 & 1 \end{bmatrix}$. The corners go to

$$(0, 0) \mapsto (0, 0), \quad (1, 0) \mapsto (1, 0.5), \quad (1, 1) \mapsto (1, 1.5), \quad (0, 1) \mapsto (0, 1).$$

The left side stays on the y -axis and the right side is lifted by 0.5: a parallelogram with two vertical sides, tilted upward to the right. Every vertical line keeps its length, since the whole line is lifted by the same amount.

Exercise 4.4

Name each transformation, reading it from its columns:

$$(a) \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \quad (d) \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \quad (e) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Solution to Exercise 4.4

- (a) $\hat{i} \mapsto \bar{0}$ and $\hat{j}$ stays: every point keeps its height and moves sideways onto the y -axis.
Projection onto the y -axis.
 (b) Both unit vectors reverse: $(x, y) \mapsto (-x, -y)$. **Rotation by 180°** (a half turn, not a mirror).
 (c) $\hat{i}$ stays and $\hat{j} \mapsto \begin{bmatrix} 0 \\ -2 \end{bmatrix}$: heights are reversed and doubled. A **flip across the x -axis combined with a vertical stretch by 2**, which is a scaling with a negative factor.

- (d) $\hat{i} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $\hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$: $(x, y) \mapsto (-y, -x)$. **Reflection across the line $y = -x$** , built in Section 3.2.
- (e) Nothing moves. The **identity**.

Exercise 4.5

A dataset records each patient's (weight in pounds, height in inches). Using $1 \text{ lb} \approx 0.4536 \text{ kg}$ and $1 \text{ in} = 2.54 \text{ cm}$, write the matrix that converts a patient to (kg, cm), and apply it to a patient at (176, 68). Is this conversion a linear transformation? Would converting temperatures from Fahrenheit to Celsius be one?

Solution to Exercise 4.5

Each coordinate is multiplied by its own factor, so the conversion is the diagonal matrix

$$\begin{bmatrix} 0.4536 & 0 \\ 0 & 2.54 \end{bmatrix} \begin{bmatrix} 176 \\ 68 \end{bmatrix} = \begin{bmatrix} 0.4536 \times 176 \\ 2.54 \times 68 \end{bmatrix} \approx \begin{bmatrix} 79.8 \\ 172.7 \end{bmatrix} :$$

about 79.8 kg and 172.7 cm. It is a linear transformation, since it is a matrix (a scaling). Converting Fahrenheit to Celsius, $C = (F - 32)/1.8$, is not: it subtracts 32 first, a translation, so 0°F does not go to 0°C and the origin moves.

5 Length-Preserving Transformations

5.1 Which Transformations Keep Lengths?

Rotate a shape and it comes out the same size; shear it and it does not. Which matrices leave every length alone? In symbols, we want

$$\|A\bar{x}\| = \|\bar{x}\| \quad \text{for every } \bar{x},$$

where $\|\bar{x}\|$ is the length (the norm) of $\bar{x}$ from Unit 2-2.

In plain words, a length-preserving matrix is a **rigid motion** that keeps the origin in place: you may pick up the plane, turn it, and flip it over, but you may not stretch it, squash it, or bend it.

Worked example: measuring before and after. Take $\bar{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, of length $\sqrt{3^2 + 4^2} = 5$, and push it through four members of the catalogue.

Matrix	$A\bar{x}$	$\ A\bar{x}\ $	Length kept?
Rotation $R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} -4 \\ 3 \end{bmatrix}$	$\sqrt{16 + 9} = 5$	yes
Shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 7 \\ 4 \end{bmatrix}$	$\sqrt{49 + 16} = \sqrt{65} \approx 8.06$	no
Scaling $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$	$\begin{bmatrix} 6 \\ 2 \end{bmatrix}$	$\sqrt{36 + 4} = \sqrt{40} \approx 6.32$	no
Projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 0 \end{bmatrix}$	3	no

Only the rotation kept the length. A single vector whose length changes is enough to prove a matrix does not preserve length. The reverse is not true, as the box below explains, and that is why we want an algebraic test that covers every vector at once.

Two facts from earlier units provide that test. First, the squared length of a vector is its dot product with itself, and a dot product can be written as a row times a column: $\|\vec{v}\|^2 = \vec{v}^\top \vec{v}$. Second, the transpose of a product reverses the order: $(AB)^\top = B^\top A^\top$ (Unit 2-3). Apply the first fact to the vector $A\vec{x}$, then the second:

$$\|A\vec{x}\|^2 = (A\vec{x})^\top (A\vec{x}) = \vec{x}^\top A^\top A \vec{x}.$$

If $A^\top A = I$, the middle of that expression is just I , and since $I\vec{x} = \vec{x}$ it reduces to $\vec{x}^\top \vec{x} = \|\vec{x}\|^2$. So every length is preserved.

Definition: Orthogonal matrix

A square matrix A with

$$A^\top A = I$$

is called **orthogonal**. Every orthogonal matrix preserves length: $\|A\vec{x}\| = \|\vec{x}\|$ for every $\vec{x}$.

What the test says about the columns. By the row-by-column rule, the (i, j) entry of $A^\top A$ is row i of $A^\top$ dotted with column j of A . But row i of $A^\top$ is column i of A , so the (i, j) entry of $A^\top A$ is **column i dotted with column j** . For a 2×2 matrix with columns $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$,

$$A^\top A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix}.$$

Setting this equal to I asks for two things:

- the diagonal entries are 1: each column dotted with itself is 1, so **each column has length 1**;
- the off-diagonal entries are 0: the two columns have dot product 0, so they are **perpendicular**.

Unit vectors that are mutually perpendicular form an **orthonormal basis**, the nicest kind of basis from Unit 2-2. Geometrically, an orthogonal matrix moves the standard axes onto another set of perpendicular unit axes.

Check on $R(\theta)$. The columns are $\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ and $\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$. Their lengths are

$$\sqrt{\cos^2 \theta + \sin^2 \theta} = 1,$$

because $(\cos \theta, \sin \theta)$ is a point on the circle of radius 1, and their dot product is

$$(\cos \theta)(-\sin \theta) + (\sin \theta)(\cos \theta) = 0.$$

So every rotation is orthogonal, whatever the angle.

Notebook U2-4_Transformations-2_NamedTransforms.ipynb, §2, lengths measured after each matrix

Worked example: the test on two reflections. For the flip across the x -axis, $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, the transpose is F itself (it is symmetric), and

$$F^\top F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

For the mirror at 30° from Section 4.3, use the column test. The columns $\begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$ and $\begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$ each have length $\sqrt{\frac{1}{4} + \frac{3}{4}} = 1$, and their dot product is $\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \cdot \left(-\frac{1}{2}\right) = 0$. So $M_{30}^\top M_{30} = I$ as well. Every mirror through the origin keeps lengths, as a mirror should.

Worked example: the scan matrix is not orthogonal. For $A = 1.5 R(30^\circ)$, the scalar comes out of the transpose, so

$$A^\top A = (1.5)^2 R(30^\circ)^\top R(30^\circ) = 2.25 I \neq I.$$

Putting this into the length formula gives $\|A\bar{x}\|^2 = \bar{x}^\top (2.25 I) \bar{x} = 2.25 \|\bar{x}\|^2$, so $\|A\bar{x}\| = 1.5 \|\bar{x}\|$: every length is multiplied by exactly 1.5. That is not a flaw, it is what the radiologist asked for, and it matches the pixel that moved from distance 100 to distance 150 in Section 4.2. The algebra did more than say “not orthogonal”: it told us precisely how lengths change.

Worked example (harder): unit columns are not enough. Consider

$$C = \begin{bmatrix} 1 & 0.6 \\ 0 & 0.8 \end{bmatrix}.$$

Both columns have length 1: the first is $\hat{i}$, and the second has length $\sqrt{0.36 + 0.64} = 1$. But their dot product is $1 \cdot 0.6 + 0 \cdot 0.8 = 0.6$, not 0, so

$$C^\top C = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix} \neq I.$$

To see a length actually change, try $\bar{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, of length $\sqrt{2} \approx 1.414$:

$$C \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.4 \\ -0.8 \end{bmatrix}, \quad \|C\bar{x}\| = \sqrt{0.16 + 0.64} = \sqrt{0.8} \approx 0.894.$$

The vector got shorter. Unit-length columns keep $\hat{i}$ and $\hat{j}$ the right length, but because the columns lean toward each other, combinations of them do not come out right. Both conditions are needed.

Non-example: perpendicular columns are not enough either. The matrix $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ from Exercise 3.1 has perpendicular columns (dot product $-1 + 1 = 0$), but each has length $\sqrt{2}$, and $A^\top A = 2I$. By the length formula every length is multiplied by $\sqrt{2}$, which is exactly the “turns by 45° and stretches by $\sqrt{2}$ ” found in that exercise.

Watch out

Two traps with this test.

- **One vector proves nothing in the positive direction.** The shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ sends $\hat{i}$ to itself, so $\hat{i}$ keeps its length, yet the shear stretches $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ from 5 to about 8.06. A matrix passes only if *every* vector keeps its length, which is what $A^\top A = I$ guarantees. One failure is enough to disqualify; one success is not enough to qualify.
- **The name is misleading.** An “orthogonal matrix” needs *orthonormal* columns: perpendicular **and** of length 1. Perpendicular columns alone, as in $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, do not make a matrix orthogonal. The name is a historical accident that everyone now has to live with.

5.2 Sorting the Catalogue

The column test sorts the catalogue of Section 4 into two groups.

Orthogonal (length-preserving): rotations, reflections, and permutations. In each of these matrices both columns are unit vectors and the two are perpendicular, so $A^\top A = I$.

Angles are preserved too. The angle between two vectors is computed from their dot product (Unit 2-2), and after the move the dot product is

$$(A\bar{u})^\top (A\bar{v}) = \bar{u}^\top A^\top A \bar{v} = \bar{u}^\top \bar{v},$$

exactly the dot product before the move. With lengths and angles both unchanged, shapes come out **congruent**, meaning the same shape and the same size: turned or flipped, never distorted.

Not orthogonal: scaling, shear, and projection.

- **Scaling** $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$: $A^\top A = \begin{bmatrix} 4 & 0 \\ 0 & 1/4 \end{bmatrix} \neq I$. Lengths along the x -axis double (and lengths along the y -axis halve).
- **Shear** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: the second column $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ has length $\sqrt{2}$, not 1. In the picture, the F's upright stem, which ran from $(0,0)$ to $(0,1)$, now runs from $(0,0)$ to $(1,1)$ and is longer.
- **Projection** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$: the second column is $\bar{0}$. Some vectors lose their length entirely; $\hat{j}$ itself is sent to $\bar{0}$.

Here is the whole sorting in one table, including the extra members met in these notes. Every entry of the middle column is a 2×2 product you can check by hand.

Matrix	$A^\top A$	Orthogonal?
any rotation $R(\theta)$	I	yes
flip across the x -axis $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	I	yes
flip across $y = -x$, $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$	I	yes
mirror at 30° , M_{30}	I	yes
permutation $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	I	yes
scaling $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$	$\begin{bmatrix} 4 & 0 \\ 0 & 1/4 \end{bmatrix}$	no
uniform enlargement $1.5 I$	$2.25 I$	no
shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$	no
projection onto the x -axis $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	no
projection onto $y = x$, P_{diag}	P_{diag}	no

Why data science cares. PCA's change of basis is orthogonal, so distances between data points are exactly what they were. The distance between two points $\bar{p}$ and $\bar{q}$ is the length $\|\bar{p} - \bar{q}\|$ (Unit 2-2); after the change of basis it is $\|A\bar{p} - A\bar{q}\| = \|A(\bar{p} - \bar{q})\|$, and an orthogonal A leaves that equal to $\|\bar{p} - \bar{q}\|$. The new coordinates are a rotation of the old ones, not a distortion.

Worked example: who is the nearest patient? Three patients are plotted by two standardized lab values: $\bar{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\bar{q} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$, $\bar{r} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. Which of q and r is most similar to patient p , measured by distance? Now transform all three with the rotation $R(90^\circ)$, and separately with the scaling $S = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$, and ask again.

	original	after $R(90^\circ)$	after S
patient p	$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
patient q	$\begin{bmatrix} 4 \\ 6 \end{bmatrix}$	$\begin{bmatrix} -6 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 8 \\ 3 \end{bmatrix}$
patient r	$\begin{bmatrix} 5 \\ 2 \end{bmatrix}$	$\begin{bmatrix} -2 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 10 \\ 1 \end{bmatrix}$
distance p to q	$\ \begin{bmatrix} -3 \\ -4 \end{bmatrix}\ = 5$	$\ \begin{bmatrix} 4 \\ -3 \end{bmatrix}\ = 5$	$\ \begin{bmatrix} -6 \\ -2 \end{bmatrix}\ = \sqrt{40} \approx 6.32$
distance p to r	$\ \begin{bmatrix} -4 \\ 0 \end{bmatrix}\ = 4$	$\ \begin{bmatrix} 0 \\ -4 \end{bmatrix}\ = 4$	$\ \begin{bmatrix} -8 \\ 0 \end{bmatrix}\ = 8$
nearest to p	r	r	q

The rotation moved every patient but left both distances exactly as they were, so r is still nearest. The scaling doubled the first lab value and halved the second. The gap from p to r lies entirely along the first value, so it doubled, from 4 to 8; the gap from p to q lies mostly along the second value, which was halved, so it grew only to about 6.32. Now q is nearest. A k -nearest-neighbors classifier, which labels a new patient by the most similar patients on file, would change its answer after the rescaling and not after the rotation. That is the practical meaning of “orthogonal matrices preserve distance,” and one reason the choice of units and scales matters so much in data science.

Worked example: angles before and after. The vectors $\bar{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ make a 45° angle. The shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ sends them to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, whose angle has

$$\cos(\text{angle}) = \frac{\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}}{1 \cdot \sqrt{5}} = \frac{2}{\sqrt{5}} \approx 0.894,$$

an angle of about 26.6° : the shear squeezed the angle. For a rotation there is nothing to compute: by the identity above, the dot product, and both lengths, are unchanged, so the cosine is still $\frac{1}{\sqrt{2}}$ and the angle is still 45° . Orthogonal matrices keep angles; other matrices usually do not.

Now test some matrices yourself.

Exercise 5.1

For each matrix, compute $A^\top A$ and say whether the transformation preserves length. For the ones that do, say what they do geometrically.

$$A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

Then: without any computation, is $2A_1$ orthogonal?

Solution to Exercise 5.1

1. Transpose A_1 by turning its rows into columns, then multiply row by column:

$$A_1^\top A_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 + 1 \cdot 1 & 0 \cdot (-1) + 1 \cdot 0 \\ (-1) \cdot 0 + 0 \cdot 1 & (-1)(-1) + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

So $A_1^\top A_1 = I$: A_1 is **orthogonal**. It is the quarter-turn rotation $R(90^\circ)$.

2. In the same way,

$$A_2^\top A_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \neq I.$$

A_2 is **not orthogonal**. Read the entries as column facts: the 2 in the corner is the squared length of the second column, so the shear stretches that column to length $\sqrt{2}$; and the off-diagonal 1s say the two columns are not even perpendicular.

3. Use the column test. Both columns have length

$$\sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{1} = 1,$$

and their dot product is $\frac{3}{5} \cdot \left(-\frac{4}{5}\right) + \frac{4}{5} \cdot \frac{3}{5} = -\frac{12}{25} + \frac{12}{25} = 0$. So $A_3^\top A_3 = I$: A_3 is **orthogonal**.

Comparing with $R(\theta)$, it is a rotation with $\cos \theta = \frac{3}{5}$ and $\sin \theta = \frac{4}{5}$, which is a turn of about 53° . (The 3-4-5 triangle again: $3^2 + 4^2 = 5^2$ is exactly why the columns come out with length 1.)

4. **No.** $2A_1$ is the quarter turn followed by doubling, and doubling changes lengths: $2A_1$ sends $\hat{i}$ to $2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, a vector of length 2. Algebraically,

$$(2A_1)^\top (2A_1) = 4A_1^\top A_1 = 4I \neq I.$$

The same reasoning gives $(kA)^\top (kA) = k^2 I$ for any orthogonal A , which equals I only when $k^2 = 1$. So scaling an orthogonal matrix by any $k \neq \pm 1$ breaks the property.

5.3 More Practice: Orthogonal or Not?

Exercise 5.2

Decide which of these are orthogonal, and describe what each orthogonal one does.

$$B_1 = \begin{bmatrix} 0.6 & 0.8 \\ 0.8 & -0.6 \end{bmatrix}, \quad B_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}, \quad B_4 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

For B_1 , check what it does to $(2, 1)$ and to $(-1, 2)$.

Solution to Exercise 5.2

B_1 : The columns $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ and $\begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix}$ have length $\sqrt{0.36 + 0.64} = 1$ each, and their dot product is $0.48 - 0.48 = 0$. So $B_1^\top B_1 = I$: **orthogonal**. It is not of the rotation form $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ (its bottom-right entry has the wrong sign), and the two checks show what it is:

$$B_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.2 + 0.8 \\ 1.6 - 0.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad B_1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.6 + 1.6 \\ -0.8 - 1.2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

The first vector stays put and the second, which is perpendicular to it, is reversed. Points along $(2, 1)$ are fixed and the perpendicular direction is flipped: B_1 is the **reflection across the line through $(2, 1)$** .

B_2 : The columns $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ have length $\sqrt{\frac{1}{2} + \frac{1}{2}} = 1$ and dot product $\frac{1}{2}(1 - 1) = 0$: **orthogonal**. It has the form of $R(-\theta)$ with $\cos \theta = \sin \theta = \frac{1}{\sqrt{2}}$, so it is the **rotation clockwise by 45°** .

B_3 : $B_3^\top B_3 = \begin{bmatrix} 0.25 & 0 \\ 0 & 4 \end{bmatrix} \neq I$: **not orthogonal**. It is a scaling that halves widths and doubles heights.

B_4 : Its columns are $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$: the three standard unit vectors in a new order. Each has length 1, and any two of them are perpendicular, so $B_4^\top B_4 = I$ (the 3×3 identity): **orthogonal**. It is the vital-signs permutation from Section 4.3, and reordering the entries of a vector never changes its length.

Exercise 5.3

Fill in the second column so that $\begin{bmatrix} 0.8 & ? \\ 0.6 & ? \end{bmatrix}$ is orthogonal. How many answers are there, and what does each one do?

Solution to Exercise 5.3

The second column must have length 1 and be perpendicular to $\begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$. Swapping the two entries and negating one of them does both: $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$ and $\begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$ each have length 1, and each has dot product 0 with $\begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$. These are the only two answers, because in the plane there are exactly two unit vectors perpendicular to a given one, pointing in opposite directions.

- $\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$ has the rotation form $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ with $\cos \theta = 0.8$ and $\sin \theta = 0.6$: a **rotation** by about 36.9° .
- $\begin{bmatrix} 0.8 & 0.6 \\ 0.6 & -0.8 \end{bmatrix}$ is a **reflection**: it sends $\hat{j}$ to the opposite of where the rotation sends it, so the F comes out mirror-reversed.

Every 2×2 orthogonal matrix is one of these two kinds, a rotation or a reflection.

Exercise 5.4

Patients $\bar{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\bar{q} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ are 5 apart. Transform both with $A_3 = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$ from Exercise 5.1 and confirm the distance is still 5.

Solution to Exercise 5.4

Column by column,

$$A_3\bar{p} = 1 \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} + 2 \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad A_3\bar{q} = 4 \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} + 6 \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} = \begin{bmatrix} -12/5 \\ 34/5 \end{bmatrix} = \begin{bmatrix} -2.4 \\ 6.8 \end{bmatrix}.$$

The difference is $A_3\bar{p} - A_3\bar{q} = \begin{bmatrix} 1.4 \\ -4.8 \end{bmatrix}$, and its length is $\sqrt{1.96 + 23.04} = \sqrt{25} = 5$. The two patients were both moved (turned by about 53°), but the distance between them did not change, exactly as an orthogonal matrix promises.

6 Composition and Order

6.1 Two Moves in a Row Is One Matrix

Doing one function and then another is called **composition**. For matrices, composition is a product.

Composition is the product

Apply B first, then A : the point $\bar{x}$ goes to $B\bar{x}$, and then to $A(B\bar{x}) = (AB)\bar{x}$. So the two-step move is the **single matrix** AB , with the **first** move on the **right**, next to $\bar{x}$.

The step $A(B\bar{x}) = (AB)\bar{x}$ is the associative rule for products from Unit 2-3. The order may look backwards at first, but it works the same way as for ordinary functions: in $f(g(x))$, the function g acts first because it sits next to x .

Worked example. Take the quarter turn and the flip across the x -axis,

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

and combine them in both orders. Since each column of a product is the left matrix times the matching column of the right one (Unit 2-3), we can build each product by following $\hat{i}$ and $\hat{j}$ through both moves.

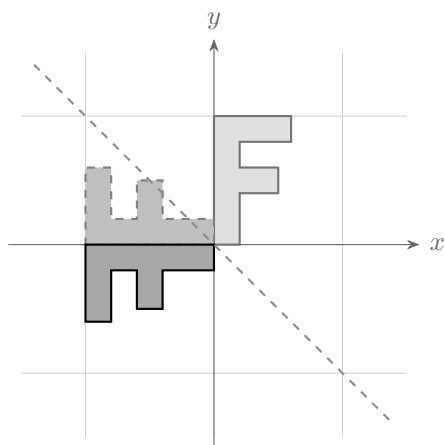
- **Rotate, then flip** is FR (R acts first, so it sits on the right). R sends $\hat{i}$ to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and F then flips that to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$. R sends $\hat{j}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, which F leaves alone.
- **Flip, then rotate** is RF . F leaves $\hat{i}$ alone, and R turns it to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. F sends $\hat{j}$ to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$, and R turns that to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Stacking the images as columns,

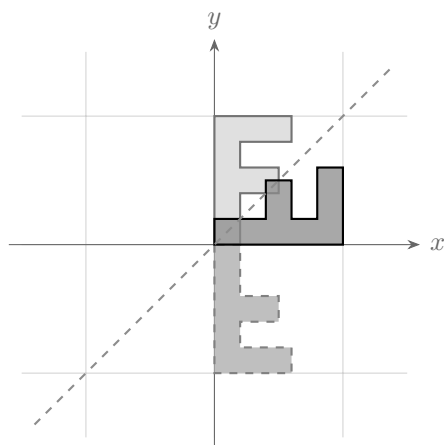
$$\text{rotate then flip: } FR = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \quad \text{flip then rotate: } RF = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

They differ, and the columns say how. RF is the permutation from Section 4: it swaps the coordinates, a reflection across $y = x$. FR sends $\hat{i} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, so (x, y) goes to $(-y, -x)$: a reflection across $y = -x$, the very matrix we built by hand in Section 3.2. This is $AB \neq BA$ from Unit 2-3, now visible as a fact about the world rather than a quirk of arithmetic.

The pictures show both orders. In each, the light F is the start, the dashed F is the shape after the first move, and the dark F is the shape after both.



Rotate, then flip: FR . Light: start. Dashed: after R . Dark: after F . The result is the start reflected across $y = -x$.



Flip, then rotate: RF . Light: start. Dashed: after F . Dark: after R . The result is the start reflected across $y = x$.

Same two moves, opposite order, different final position. Read the dark F's columns off either picture (where its bottom edge and its stem now point) and you recover the product computed above.

Notebook U2-4_Transformations-2_NamedTransforms.ipynb, §3, both orders side by side

Here are more compositions, each built the same way: follow $\hat{i}$ and $\hat{j}$ through the moves in the order they happen, or multiply the matrices with the first move on the right.

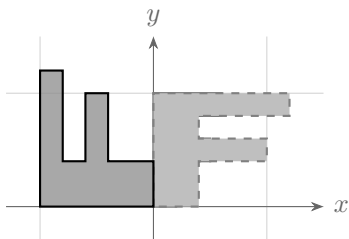
Worked example: stretch and turn. Let $S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ double every width, and let $R = R(90^\circ)$ as before.

- **Stretch, then turn** is RS . S sends $\hat{i}$ to $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, and R turns that up to $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$. S leaves $\hat{j}$ alone, and R turns it to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$. So $RS = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$.
- **Turn, then stretch** is SR . R turns $\hat{i}$ up to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, which S leaves alone (it has no width to double). R turns $\hat{j}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, which S doubles to $\begin{bmatrix} -2 \\ 0 \end{bmatrix}$. So $SR = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$.

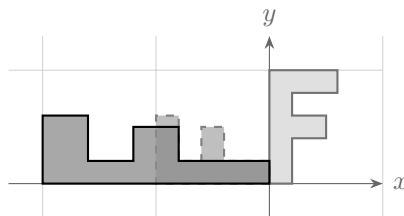
Multiplying the matrices row by column gives the same results:

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}, \quad SR = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}.$$

On the point $(1,1)$: RS gives $(-1,2)$ and SR gives $(-2,1)$. The difference has a plain meaning. Stretching *before* turning doubles the F's bars, which are horizontal at the start; the turn then stands them upright. Stretching *after* turning doubles whatever is horizontal at the end, which by then is the F's stem, lying on its side. One order lengthens the bars, the other lengthens the stem, as the pictures show.



Stretch, then turn: RS . Dashed: after S .
Dark: after R .

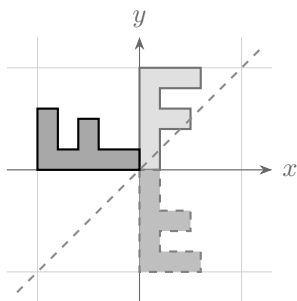


Turn, then stretch: SR . Dashed: after R . Dark:
after S .

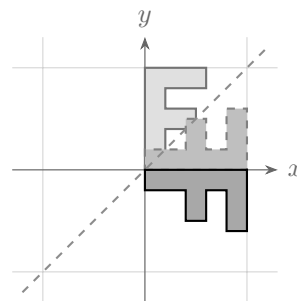
Worked example: two reflections make a rotation. Take the flip $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ across the x -axis and the swap $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the flip across $y = x$.

- **Flip, then swap** is PF . F leaves $\hat{i}$ alone and P sends it to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$; F sends $\hat{j}$ to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and P sends that to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$. So $PF = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = R(90^\circ)$.
- **Swap, then flip** is FP . P sends $\hat{i}$ to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and F flips it to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$; P sends $\hat{j}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, which F leaves alone. So $FP = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = R(-90^\circ)$.

Two mirror flips cancel each other's mirroring, so what is left is a turn, and the order decides which way it turns. In the pictures the dark F is readable (not mirror-reversed), only turned.



Flip, then swap: $PF = R(90^\circ)$. Dashed: after
 F .



Swap, then flip: $FP = R(-90^\circ)$. Dashed:
after P .

Worked example: rotations add their angles. Turning by 30° and then by 60° should be a turn by 90° . Check with exact values:

$$R(60^\circ)R(30^\circ) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & -\frac{1}{4} - \frac{3}{4} \\ \frac{3}{4} + \frac{1}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = R(90^\circ).$$

Multiplying in the other order gives $R(90^\circ)$ as well (try it), because it does not matter which of two turns about the same point comes first. So “order matters” is the general rule, but some pairs do commute: turns with turns, and, as the exercise below shows, uniform scaling with anything.

Worked example: three moves. Shear with $H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then turn with $R = R(90^\circ)$, then double with $2I$. The first move goes on the right and the last on the left, so the single matrix is $(2I)RH$:

$$RH = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \quad (2I)RH = \begin{bmatrix} 0 & -2 \\ 2 & 2 \end{bmatrix}.$$

Check on $(1, 1)$ one step at a time: the shear gives $(2, 1)$, the turn gives $(-1, 2)$, the doubling gives $(-2, 4)$. The single matrix gives $1 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$ in one step. Read a product from right to left and you read off the order in which the moves happen.

Worked example: why a neural network needs its nonlinear step. Section 1.1 described a network layer as a matrix followed by a simple nonlinear step. Suppose we left the nonlinear step out and stacked two layers, $W_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and then $W_2 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. The two layers together are the single matrix

$$W_2 W_1 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

On the input $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$: layer one gives $W_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, layer two gives $W_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$, and the single matrix gives $\begin{bmatrix} 1+2 \\ 3+7 \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$ directly. Two linear layers do nothing that one linear layer could not, and neither would a hundred. The nonlinear step between layers is what stops the stack from collapsing into one matrix, and it is the reason deep networks can learn things a single matrix cannot.

Watch out

The order you say it is not the order you write it. “Rotate, then flip” is FR , not RF : the first move sits on the right, next to the vector. Two ways to avoid the slip: write the vector first and add each move to its left as it happens ($\bar{x}$, then $R\bar{x}$, then $FR\bar{x}$); or, after multiplying, follow $\hat{i}$ through the moves in words and confirm that it lands on the first column of your product.

6.2 The Same Move Again and Again: Powers

Applying one transformation n times in a row is the matrix power A^n from Unit 2-3, the matrix multiplied by itself n times. The catalogue makes the powers easy to predict, and they fall into three kinds of behaviour.

- **Rotation cycles.** Four quarter turns make a full turn, which puts everything back where it started: $R(90^\circ)^4 = I$. In general, turning by θ a total of n times is turning by $n\theta$, so $R(\theta)^n = R(n\theta)$.
- **Projection is idempotent.** A matrix is **idempotent** if applying it twice does the same as applying it once. Flattening a shape that is already flat changes nothing, so $P^2 = P$, and therefore $P^n = P$ for every $n \geq 1$. Once the information is gone, it stays gone.
- **Shear and scaling accumulate.** Each application of the shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ adds the height y to x once more, so after n steps x has become $x + ny$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix},$$

which leans further each time. Likewise $(kI)^n = k^n I$: every step multiplies lengths by the same factor k , so the shape grows or shrinks geometrically.

Check by multiplying. For the shear and the projection $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

The shear squared leans twice as far, and the projection squared is the projection again. That also makes the shear a **non-example** of idempotence: its square is not the shear, because a second push moves points further than the first one did.

A reflection is its own undo: flip twice and you are back where you started, so $F^2 = I$. That makes it the first **inverse** you have met, a transformation that undoes another one. Unit 2-5 asks which transformations have one.

Worked example: following a point around the cycle. Apply $R = R(90^\circ)$ to $(2, 1)$ again and again. Each application sends (x, y) to $(-y, x)$:

n	0	1	2	3	4
$R^n \begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} -2 \\ -1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

After four steps the point is home again, and the cycle repeats: $R^5 = R$, $R^6 = R^2$, and so on. So to find R^n for a large n , only the remainder when n is divided by 4 matters.

Worked example: two eighth turns. $R(45^\circ)^2$ should be $R(90^\circ)$. With $R(45^\circ) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, the two factors of $\frac{1}{\sqrt{2}}$ multiply to $\frac{1}{2}$:

$$R(45^\circ)^2 = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1-1 & -1-1 \\ 1+1 & -1+1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = R(90^\circ).$$

It follows that $R(45^\circ)^8 = R(90^\circ)^4 = I$: eight eighth turns make a full turn.

Worked example: more powers from the catalogue.

- **Shear, cubed.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, matching the pattern $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$. It sends $(0, 1)$ to $(3, 1)$: three pushes of one unit each.
- **Projection onto $y = x$.** For $P_{\text{diag}} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$, every entry of P_{diag}^2 is $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, so $P_{\text{diag}}^2 = P_{\text{diag}}$. In words: $(3, 1)$ projects to $(2, 2)$, and $(2, 2)$, already on the line, projects to itself.
- **Shrinking.** $(0.5 I)^3 = 0.125 I$: halving three times shrinks to one eighth. And $(1.1 I)^{10} = 1.1^{10} I \approx 2.594 I$: ten enlargements of 10% each more than double the size, the same compounding as interest.
- **The mirror at 30° .** M_{30}^2 has top-left entry $\frac{1}{2} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{1}{4} + \frac{3}{4} = 1$ and top-right entry $\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \cdot \left(-\frac{1}{2}\right) = 0$; the bottom row works out the same way, so $M_{30}^2 = I$. Every reflection undoes itself, not just the ones across the axes.

Watch out

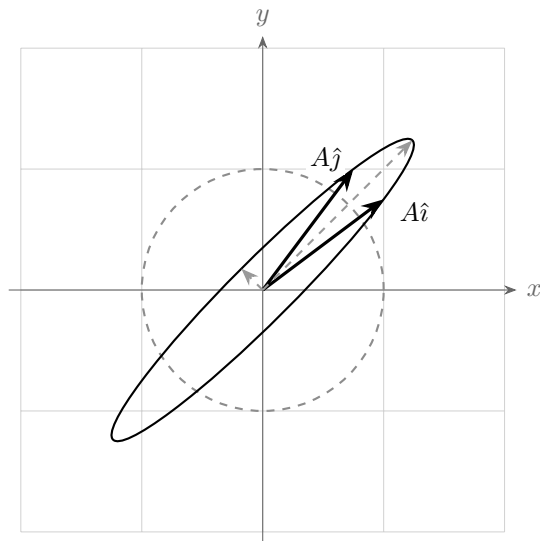
A matrix power is repeated multiplication, not squaring each entry. Squaring the entries of $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ one at a time gives $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, a swap, while the true square is $R^2 = RR = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, a half turn. The two have nothing to do with each other. The same trap is built into NumPy: for an array **A**, **A**2** squares each entry, while **A @ A** (or **np.linalg.matrix_power(A, 2)**) is the matrix power.

6.3 Looking Ahead: the Circle Becomes an Ellipse

The tips of all the unit vectors trace out the **unit circle**, the circle of radius 1 around the origin. Push every one of those unit vectors through a matrix, and the unit circle becomes an **ellipse**, a stretched circle. The figure uses

$$A = \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix}.$$

Its columns sit on the ellipse, since they are the images of $\hat{i}$ and $\hat{j}$, which sit on the unit circle.



Dashed circle: the unit circle. Solid: its image. Thin dashed arrows: the directions that do not turn.

Most arrows are **turned** as well as stretched: $\hat{i}$, for example, pointed along the x -axis but $A\hat{i}$ does not. But the arrows along the ellipse's long and short axes are **not turned at all**: they only change length. In the figure they are the thin dashed arrows, a long one along the line $y = x$ and a short one along $y = -x$.

Worked example: which arrows turn? Push four unit vectors through $A = \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix}$.

- $\hat{i} \mapsto \begin{bmatrix} 1 \\ 0.75 \end{bmatrix}$, of length $\sqrt{1 + 0.5625} = 1.25$. It was horizontal and now it is not: **turned** and stretched. The same happens to $\hat{j} \mapsto \begin{bmatrix} 0.75 \\ 1 \end{bmatrix}$.
- The unit vector $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ along $y = x$ goes to

$$A \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 + 0.75 \\ 0.75 + 1 \end{bmatrix} = 1.75 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 1.237 \\ 1.237 \end{bmatrix} :$$

exactly 1.75 times itself. **Not turned**, only stretched: this is the long dashed arrow, the tip of the ellipse.

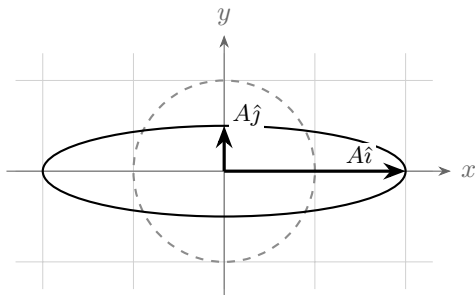
- The unit vector $\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ along $y = -x$ goes to

$$A \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 + 0.75 \\ -0.75 + 1 \end{bmatrix} = 0.25 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} -0.177 \\ 0.177 \end{bmatrix} :$$

exactly 0.25 times itself. **Not turned**, only shrunk: the short dashed arrow.

So the ellipse reaches out 1.75 along $y = x$ and only 0.25 along $y = -x$. Finding such special directions, and their stretch factors, for any matrix is exactly the job of Unit 2-6.

For the simplest matrices the answer is visible at once. A diagonal matrix such as $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$ does not turn the axes at all: it sends the unit circle to an ellipse reaching out 2 along the x -axis and $\frac{1}{2}$ along the y -axis (figure below). An orthogonal matrix keeps every length, so it sends the unit circle to the unit circle: for a rotation, the “ellipse” is the circle itself, merely turned.



The unit circle (dashed) and its image under $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$ (solid), with the columns as arrows.

This picture raises two questions that the next two units answer:

- By what factor did the **area** grow? (Unit 2-5: the determinant.)
- Which directions are **only stretched**, and by how much? (Unit 2-6: eigenvectors.)

Notebook U2-4_Transformations-3_UnitCircleAnimation.ipynb, §2, the swept vector

To finish the section, practice composition and powers.

Exercise 6.1

Let

$$R = R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

1. Write the single matrix for “double everything, then turn a quarter turn.” Then the matrix for the opposite order. Are they different? Explain why in this case.
2. Compute R^2 and say in words what it does. What is R^3 as a rotation?
3. Compute PR and RP and describe each geometrically. Which order projects onto the y -axis?

Solution to Exercise 6.1

1. “Double first, then turn” puts S on the right: RS . Its columns are R times the columns of S : $R\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ and $R\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$. “Turn first, then double” is SR , and since $S = 2I$, that is just $2R$. So

$$RS = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}, \quad SR = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}.$$

They are **the same**, because $S = 2I$ and a multiple of the identity commutes with everything: $R(2I) = 2R = (2I)R$. Geometrically, doubling does not change any direction,

so it makes no difference whether you turn before or after. Non-commutativity is the general rule, but uniform scaling is the classic exception.

- Follow $\hat{i}$ and $\hat{j}$ through two quarter turns. R sends $\hat{i}$ to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and R again sends that to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$. R sends $\hat{j}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, and R again sends that to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$. So

$$R^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I.$$

Two quarter turns make a **half turn**: every point (x, y) is sent to its opposite, $(-x, -y)$. Then $R^3 = R^2 R = -R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, three quarter turns, which is a rotation by 270° . That is the same as a quarter turn clockwise: $\hat{i}$ now points straight down.

- Build each product column by column.

- PR (turn first, then flatten): R sends $\hat{i}$ to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, which P flattens to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$; R sends $\hat{j}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, which P leaves alone. So $PR = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$: turn a quarter turn, then flatten onto the x -axis.
- RP (flatten first, then turn): P leaves $\hat{i}$ alone and R turns it to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$; P crushes $\hat{j}$ to $\bar{0}$, which stays $\bar{0}$. So $RP = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$: flatten onto the x -axis first, then turn that axis up onto the y -axis.

So RP is the one that lands everything on the y -axis. Check on $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$: P sends it to $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$, and R turns that to $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$. (By contrast, PR turns $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ to $\begin{bmatrix} -5 \\ 3 \end{bmatrix}$ and then drops it to $\begin{bmatrix} -5 \\ 0 \end{bmatrix}$ on the x -axis.) Notice that RP does not drop each point straight across onto the y -axis, which would send $(3, 5)$ to $(0, 5)$; it carries the first coordinate up onto the y -axis instead.

6.4 More Practice: Composition and Powers

Exercise 6.2

Let $H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (shear) and $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (flip across the x -axis). Compute FH and HF , say which one means “shear, then flip,” and test both on $(0, 1)$.

Solution to Exercise 6.2

“Shear, then flip” puts the shear first, on the right: FH . Row by column,

$$FH = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, \quad HF = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}.$$

On $(0, 1)$, which is $\hat{j}$, each product gives its second column: FH sends it to $(1, -1)$ and HF to $(-1, -1)$. Step by step: shearing $(0, 1)$ pushes it to $(1, 1)$ and flipping gives $(1, -1)$; flipping first gives $(0, -1)$, and the shear then pushes that point, which is at height -1 , one unit to the *left*, to $(-1, -1)$. The flip reversed the height, so the shear pushed the other way.

Exercise 6.3

Show that $R(30^\circ)R(-30^\circ) = I$, and say in words what that means.

Solution to Exercise 6.3

Write $c = \cos 30^\circ = \frac{\sqrt{3}}{2}$ and $s = \sin 30^\circ = \frac{1}{2}$. Then $R(30^\circ) = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and, from the clockwise formula in Section 4.2, $R(-30^\circ) = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$. Row by column,

$$R(30^\circ)R(-30^\circ) = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \begin{bmatrix} c^2 + s^2 & cs - sc \\ sc - cs & s^2 + c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

since $c^2 + s^2 = \frac{3}{4} + \frac{1}{4} = 1$ and the off-diagonal terms cancel. In words: turn 30° clockwise and then 30° counterclockwise, and every point is back where it started, so the combined move is the identity. Each turn undoes the other. Unit 2-5 calls a matrix that undoes another its **inverse**; here you have found one without any new machinery.

Exercise 6.4

Compute $R(60^\circ)^2$ with exact values and identify it as a rotation. Then, without multiplying, find $R(60^\circ)^3$ and $R(60^\circ)^6$.

Solution to Exercise 6.4

With $R(60^\circ) = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$,

$$R(60^\circ)^2 = \begin{bmatrix} \frac{1}{4} - \frac{3}{4} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & -\frac{3}{4} + \frac{1}{4} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}.$$

Its first column $\begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$ is $\begin{bmatrix} \cos 120^\circ \\ \sin 120^\circ \end{bmatrix}$, so this is $R(120^\circ)$: two turns of 60° make one turn of 120° , as $R(\theta)^n = R(n\theta)$ predicts. By the same rule, $R(60^\circ)^3 = R(180^\circ) = -I$, a half turn, and $R(60^\circ)^6 = R(360^\circ) = I$, a full turn that puts everything back.

Exercise 6.5

Two linear network layers, with no nonlinear step between them, apply $W_1 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ and then $W_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Find the single matrix that does both, and check it on the input $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Solution to Exercise 6.5

W_1 acts first, so it goes on the right:

$$W_2 W_1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + (-1) \cdot 1 & 1 \cdot 0 + (-1) \cdot 1 \\ 0 \cdot 2 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Check on $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in two steps: $W_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, then $W_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. In one step: $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1-2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. The same. The single matrix is the one from Exercise 3.1, which turns by 45° and stretches by $\sqrt{2}$: two layers that each shear and stretch combine into one turn-and-stretch, and nothing more.

7 Review

What we covered

1. A matrix is a **function** on vectors, $T(\bar{x}) = A\bar{x}$: it moves every point of the plane by the same rule.
2. The rule is **linear**: $A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}$ and $A(c\bar{u}) = cA\bar{u}$. So the origin is fixed, lines stay lines, and grids stay evenly spaced. Translation is not linear.
3. The same product also **changes coordinates**: $B\bar{c}$ turns $\bar{b}$ -basis coordinates into standard ones.
4. **The columns of A are where $\hat{i}$ and $\hat{j}$ land.** Read any matrix as a picture; build any transformation by choosing two images and stacking them.
5. The **catalogue**: scaling (diagonal), shear, rotation $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, reflection, permutation, and projection, which destroys information.
6. A transformation **preserves length** exactly when $A^\top A = I$ (**orthogonal**): its columns are an orthonormal basis. Rotations, reflections, and permutations qualify; scaling, shear, and projection do not.
7. **Two moves in a row are the product AB** , first move on the right. Order matters: rotate-then-flip and flip-then-rotate are different reflections.
8. **Powers** repeat a move: rotations cycle, a projection is idempotent ($P^2 = P$), shears and scalings accumulate, and a reflection undoes itself.
9. A matrix sends the unit circle to an **ellipse**. How much the area grows is the **determinant** (Unit 2-5); the directions that are only stretched are the **eigenvectors** (Unit 2-6).

Notebooks:

- U2-4_Transformations-1_MatrixAsMap.ipynb
- U2-4_Transformations-2_NamedTransforms.ipynb
- U2-4_Transformations-3_UnitCircleAnimation.ipynb

Previous: Unit 2-3, **Matrices**.

Next: Unit 2-5, **Determinants and Inverses**: how much a transformation stretches area, and when a transformation can be undone.